

**RANCANG BANGUN APLIKASI SURVEI PENGGUNA LULUSAN
BERBASIS WEB DENGAN DASHBOARD INTERAKTIF
MENGUNAKAN METODE PIECES**

LAMPIRAN SOURCE CODE UTAMA

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1. Routing Utama Aplikasi

File path: routes/web.php

Definisi rute aplikasi yang menghubungkan autentikasi, dashboard, pengelolaan survei, pengisian responden, dan laporan.

```

1  <?php
2
3  use App\Http\Controllers\DashboardController;
4  use App\Http\Controllers\LulusanController;
5  use App\Http\Controllers\PenggunaLulusanController;
6  use App\Http\Controllers\PertanyaanController;
7  use App\Http\Controllers\ProfileController;
8  use App\Http\Controllers\ReportController;
9  use App\Http\Controllers\SurveyController;
10 use Illuminate\Support\Facades\Route;
11
12 Route::get('/', function () {
13     return view('landing');
14 }->name('landing');
15
16 Route::get('/user-manual', function () {
17     return response()->file(base_path('docs/user-manual-role-user.pdf'));
18 }->name('user-manual');
19
20 Route::post('/access-survey', [SurveyController::class, 'verifyCode']->name('survey.access'));
21 Route::get('/fill-survey/{code}', [SurveyController::class, 'fill']->name('survey.fill'));
22 Route::post('/submit-survey/{code}', [SurveyController::class, 'submitJawaban']->name('survey.submit'));
23
24 Route::middleware('auth')->group(function () {
25
26     // --- AKSES SEMUA ROLE (Admin & User Regular) ---
27     Route::get('/dashboard', [DashboardController::class, 'index']->name('dashboard'));
28
29     Route::get('/profile', [ProfileController::class, 'edit']->name('profile.edit'));
30     Route::patch('/profile', [ProfileController::class, 'update']->name('profile.update'));
31     Route::delete('/profile', [ProfileController::class, 'destroy']->name('profile.destroy'));
32
33
34
35     // --- KHUSUS ROLE ADMIN ---
36     // Ganti 'admin' sesuai dengan 'code' yang ada di tabel roles Anda
37     Route::middleware(['role:admin']->group(function () {
38
39         // Survey
40         Route::get('/survey', [SurveyController::class, 'index']->name('survey'));
41         Route::get('/addsurvey', [SurveyController::class, 'add']->name('addsurvey'));
42         Route::post('/survey/store', [SurveyController::class, 'store']->name('survey.store'));
43         Route::get('/survey/bulk', [SurveyController::class, 'bulkCreate']->name('survey.bulk'));
44         Route::post('/survey/bulk', [SurveyController::class, 'bulkStore']->name('survey.bulk.store'));
45         Route::get('/survey/lulusan-by-tahun', [SurveyController::class, 'getLulusanByTahun']->name('survey.lulusan-by-tahun'));
46         Route::get('/get-perusahaan/{id}', [SurveyController::class, 'getPerusahaanData']);

```

```

47 Route::get('/survey/{id}/edit', [SurveyController::class, 'edit'])->name('survey.edit');
48 Route::put('/survey/{id}', [SurveyController::class, 'update'])->name('survey.update');
49 Route::delete('/survey/{id}', [SurveyController::class, 'destroy'])->name('survey.destroy');
50
51 // Lulusan
52 Route::get('/lulusan', [LulusanController::class, 'index'])->name('lulusan');
53 Route::get('/addgrad', [LulusanController::class, 'add'])->name('addgrad');
54 Route::post('/lulusan.store', [LulusanController::class, 'store'])->name('lulusan.store');
55 Route::get('/lulusan/{id}', [LulusanController::class, 'show'])->name('lulusan.show');
56
57 // Pengguna Lulusan
58 Route::get('/penggunalulusan', [PenggunaLulusanController::class, 'index'])->name('penggunalulusan');
59 Route::get('/create', [PenggunaLulusanController::class, 'create'])->name('create');
60 Route::post('/pengguna.store', [PenggunaLulusanController::class, 'store'])->name('pengguna.store');
61 Route::get('/penggunalulusan/{id}/edit', [PenggunaLulusanController::class, 'edit'])->name('penggunalulusan.edit');
62 Route::put('/penggunalulusan/{id}', [PenggunaLulusanController::class, 'update'])->name('penggunalulusan.update');
63 Route::delete('/penggunalulusan/{id}', [PenggunaLulusanController::class, 'destroy'])->name('penggunalulusan.destroy');
64
65 // Kategori
66 Route::resource('kategori', \App\Http\Controllers\KategoriController::class)->except(['show']);
67
68 // Report
69 Route::get('/report', [ReportController::class, 'index'])->name('report');
70 Route::get('/report/download', [ReportController::class, 'download'])->name('report.download');
71 Route::get('/report/arsip', [ReportController::class, 'arsip'])->name('report.arsip');
72 Route::get('/report/arsip/{id}', [ReportController::class, 'arsipDetail'])->name('report.arsip.detail');
73
74 // Pertanyaan
75 Route::get('/pertanyaan', [PertanyaanController::class, 'index'])->name('pertanyaan');
76 Route::get('/addquestion', [PertanyaanController::class, 'add'])->name('addquestion');
77 Route::get('/pertanyaan/{id}/edit', [PertanyaanController::class, 'edit'])->name('pertanyaan.edit');
78 Route::get('/pertanyaan/{id}/switch', [PertanyaanController::class, 'switch'])->name('pertanyaan.switch');
79 Route::put('/pertanyaan/{id}', [PertanyaanController::class, 'update'])->name('pertanyaan.update');
80 Route::post('/savequestion', [PertanyaanController::class, 'store'])->name('savequestion');
81
82 });
83 });
84
85 require __DIR__ . '/auth.php';
86

```

2. Controller Dashboard

File path: app/Http/Controllers/DashboardController.php

Controller yang menerima filter periode, fakultas, dan program studi sebelum meneruskan data terolah ke dashboard.

```
1 <?php
2
3 namespace App\Http\Controllers;
4
5 use App\Services\DashboardService;
6 use Illuminate\Http\Request;
7
8 class DashboardController extends Controller
9 {
10     protected $dashboardService;
11
12     public function __construct(DashboardService $dashboardService)
13     {
14         $this->dashboardService = $dashboardService;
15     }
16
17     public function index(Request $request)
18     {
19         $data = $this->dashboardService->getDashboardData($request->only([
20             'periode',
21             'fakultas',
22             'program_studi',
23         ]));
24
25         return view('admin.dashboard.index', $data);
26     }
27 }
28
```

3. Service Perhitungan Dashboard

File path: app/Services/DashboardService.php

Logika utama perhitungan indeks kepuasan, rata-rata kategori, distribusi jawaban, statistik responden, dan data drill-down dashboard.

```

1  <?php
2
3  namespace App\Services;
4
5  use App\Models\SurveyArsip;
6  use Illuminate\Support\Arr;
7
8  class DashboardService
9  {
10     public function getDashboardData(array $filters = []): array
11     {
12         $periode = collect(Arr::wrap($filters['periode'] ?? []))
13             ->filter(fn ($value) => filled($value))
14             ->values()
15             ->all();
16
17         $filters['periode'] = $periode;
18
19         $fakultas = $filters['fakultas'] ?? null;
20         $programStudi = $filters['program_studi'] ?? null;
21         $fakultasProdi = $this->getFakultasProdiOptions();
22
23         if ($fakultas && $programStudi && !in_array($programStudi, $fakultasProdi[$fakultas] ?? [], true)) {
24             $programStudi = null;
25             $filters['program_studi'] = null;
26         }
27
28         $arsipQuery = SurveyArsip::query();
29
30         if (!empty($periode)) {
31             $arsipQuery->whereIn('tahun_instrumen', $periode);
32         }
33
34         if ($fakultas) {
35             $arsipQuery->where('lulusan_fakultas', $fakultas);
36         }
37
38         if ($programStudi) {
39             $arsipQuery->where('lulusan_program_studi', $programStudi);
40         }
41
42         $arsipList = $arsipQuery->get();
43         $totalSurvey = $arsipList->count();
44         $totalLulusan = $arsipList->pluck('lulusan_nim')->filter()->unique()->count();
45
46         $ratingByKategori = [];

```

```

47 $allRatings = [];
48
49 foreach ($arsipList as $arsip) {
50     foreach ($arsip->jawaban_json ?? [] as $item) {
51         if (($item['jenis'] ?? '') !== 'rating' || !isset($item['nilai']) || $item['nilai'] === null) {
52             continue;
53         }
54
55         $kategori = $item['kategori'] ?? 'Lainnya';
56         $nilai = (int) $item['nilai'];
57
58         $ratingByKategori[$kategori][] = $nilai;
59         $allRatings[] = $nilai;
60     }
61 }
62
63 $rataKeseluruhan = count($allRatings) > 0
64     ? array_sum($allRatings) / count($allRatings)
65     : null;
66
67 $kategoriStats = collect($ratingByKategori)
68     ->map(fn ($nilai, $kategori) => (object) [
69         'kategori' => $kategori,
70         'rata_rata' => array_sum($nilai) / count($nilai),
71         'total_respon' => count($nilai),
72     ])
73     ->sortByDesc('rata_rata')
74     ->values();
75
76 $chartLabels = $kategoriStats->pluck('kategori')->toArray();
77 $chartData = $kategoriStats->pluck('rata_rata')->map(fn ($value) => round($value, 2))->toArray();
78 $kategoriTerbaik = $kategoriStats->first();
79 $kategoriTerlemah = $kategoriStats->last();
80
81 $kepuasanPerKategori = collect($ratingByKategori)
82     ->map(function ($nilai, $kategori) {
83         $total = count($nilai);
84         $countSangatBaik = count(array_filter($nilai, fn ($value) => $value == 4));
85         $countBaik = count(array_filter($nilai, fn ($value) => $value == 3));
86         $countKurang = count(array_filter($nilai, fn ($value) => $value == 2));
87         $countSangatKurang = count(array_filter($nilai, fn ($value) => $value == 1));
88
89         return [
90             'kategori' => $kategori,
91             'pct_sb' => $total > 0 ? round($countSangatBaik / $total * 100, 1) : 0,
92             'pct_b' => $total > 0 ? round($countBaik / $total * 100, 1) : 0,
93             'pct_k' => $total > 0 ? round($countKurang / $total * 100, 1) : 0,
94             'pct_sk' => $total > 0 ? round($countSangatKurang / $total * 100, 1) : 0,
95         ];
96     })
97     ->sortKeys()
98     ->values();
99

```



```

100 $kategoriDetails = collect($ratingByKategori)
101   ->map(function ($nilai, $kategori) {
102     $total = count($nilai);
103     $counts = [
104       'sb' => count(array_filter($nilai, fn ($value) => $value == 4)),
105       'b' => count(array_filter($nilai, fn ($value) => $value == 3)),
106       'k' => count(array_filter($nilai, fn ($value) => $value == 2)),
107       'sk' => count(array_filter($nilai, fn ($value) => $value == 1)),
108     ];
109
110     return [
111       'kategori' => $kategori,
112       'rata_rata' => $total > 0 ? round(array_sum($nilai) / $total, 2) : 0,
113       'total_respon' => $total,
114       'counts' => $counts,
115       'percentages' => [
116         'sb' => $total > 0 ? round($counts['sb'] / $total * 100, 1) : 0,
117         'b' => $total > 0 ? round($counts['b'] / $total * 100, 1) : 0,
118         'k' => $total > 0 ? round($counts['k'] / $total * 100, 1) : 0,
119         'sk' => $total > 0 ? round($counts['sk'] / $total * 100, 1) : 0,
120       ],
121     ];
122   })
123   ->sortBy('kategori')
124   ->values();
125
126 $countKategori = $kepuasanPerKategori->count();
127 $sumSangatBaik = $kepuasanPerKategori->sum('pct_sb');
128 $sumBaik = $kepuasanPerKategori->sum('pct_b');
129 $sumKurang = $kepuasanPerKategori->sum('pct_k');
130 $sumSangatKurang = $kepuasanPerKategori->sum('pct_sk');
131
132 $kepuasanRingkasan = [
133   'total' => [
134     'sb' => round($sumSangatBaik, 1),
135     'b' => round($sumBaik, 1),
136     'k' => round($sumKurang, 1),
137     'sk' => round($sumSangatKurang, 1),
138   ],
139   'rata' => $countKategori > 0 ? [
140     'sb' => round($sumSangatBaik / $countKategori, 1),
141     'b' => round($sumBaik / $countKategori, 1),
142     'k' => round($sumKurang / $countKategori, 1),
143     'sk' => round($sumSangatKurang / $countKategori, 1),
144   ] : ['sb' => 0, 'b' => 0, 'k' => 0, 'sk' => 0],
145 ];
146
147 $respondenProdiStats = $arsipList
148   ->groupBy(fn ($arsip) => $arsip->lulusan_program_studi ?: 'Tidak diketahui')
149   ->map(fn ($items, $prodi) => [
150     'prodi' => $prodi,
151     'total' => $items->count(),
152   ])

```

```

153     ->sortByDesc('total')
154     ->values();
155
156     $respondenProdiLabels = $respondenProdiStats->pluck('prodi')->toArray();
157     $respondenProdiData = $respondenProdiStats->pluck('total')->toArray();
158
159     $prodiDetails = $arsipList
160     ->groupBy(fn ($arsip) => $arsip->lulusan_program_studi ?: 'Tidak diketahui')
161     ->map(function ($items, $prodi) {
162         $jenisPerusahaan = $items
163         ->groupBy(fn ($arsip) => $arsip->perusahaan_jenis ?: 'Tidak diketahui')
164         ->map(fn ($group, $jenis) => [
165             'label' => $jenis,
166             'total' => $group->count(),
167         ])
168         ->sortByDesc('total')
169         ->values();
170
171         return [
172             'prodi' => $prodi,
173             'fakultas' => $items->pluck('lulusan_fakultas')->filter()->unique()->implode(', ') ?: 'Tidak diketahui',
174             'total' => $items->count(),
175             'jenis_perusahaan' => $jenisPerusahaan,
176         ];
177     })
178     ->values();
179
180     $komentarTerbaru = $arsipList
181     ->sortByDesc('submitted_at')
182     ->map(function ($arsip) {
183         $essay = collect($arsip->jawaban_json ?? [])->firstWhere('jenis', 'essay');
184
185         if (!$essay || empty($essay['jawaban'])) {
186             return null;
187         }
188
189         return (object) [
190             'jawaban_text' => $essay['jawaban'],
191             'responden' => $arsip->penyelia_nama,
192             'soal_teks' => $essay['soal'] ?? null,
193             'nama_perusahaan' => $arsip->perusahaan_nama,
194         ];
195     })
196     ->filter()
197     ->values();
198
199     $filterOptions = $this->getFilterOptions();
200
201     return compact(
202         'totalSurvey',
203         'totalLulusan',
204         'rataKeseluruhan',
205         'kategoriTerbaik',

```

```

206         'kategoriTerlemah',
207         'chartLabels',
208         'chartData',
209         'respondenProdiLabels',
210         'respondenProdiData',
211         'prodiDetails',
212         'kepuasanPerKategori',
213         'kepuasanRingkasan',
214         'kategoriDetails',
215         'komentarTerbaru',
216         'filterOptions',
217         'filters',
218     );
219 }
220
221 private function getFilterOptions(): array
222 {
223     $fakultasProdi = $this->getFakultasProdiOptions();
224     $fakultasList = collect(array_keys($fakultasProdi));
225     $prodiList = collect($fakultasProdi->flatten()->values());
226     $fakultasLabels = $this->getFakultasLabels();
227
228     $periodeList = SurveyArsip::whereNotNull('tahun_instrumen')
229         ->distinct()
230         ->orderByDesc('tahun_instrumen')
231         ->pluck('tahun_instrumen');
232
233     return compact('periodeList', 'fakultasList', 'prodiList', 'fakultasProdi', 'fakultasLabels');
234 }
235
236 private function getFakultasProdiOptions(): array
237 {
238     $fromData = SurveyArsip::query()
239         ->whereNotNull('lulusan_fakultas')
240         ->whereNotNull('lulusan_program_studi')
241         ->get(['lulusan_fakultas', 'lulusan_program_studi'])
242         ->groupBy('lulusan_fakultas')
243         ->map(fn ($items) => $items
244             ->pluck('lulusan_program_studi')
245             ->filter()
246             ->unique()
247             ->sort()
248             ->values()
249             ->all())
250         ->sortKeys()
251         ->all();
252
253     return !empty($fromData) ? $fromData : [
254         'FTI' => ['Manajemen Informatika', 'Sistem Informasi', 'Teknik Informatika'],
255         'FEB' => ['Akuntansi', 'Ekonomi Pembangunan', 'Manajemen'],
256         'FDIK' => ['Desain Komunikasi Visual', 'Ilmu Komunikasi', 'Jurnalistik'],
257     ];
258 }

```

```
259
260 private function getFakultasLabels(): array
261 {
262     return [
263         'FTI' => 'Fakultas Teknologi dan Informatika',
264         'FEB' => 'Fakultas Ekonomi dan Bisnis',
265         'FDIK' => 'Fakultas Desain dan Industri Kreatif',
266     ];
267 }
268 }
269
```

4. Tampilan Dashboard Interaktif

File path: resources/views/admin/dashboard/index.blade.php

Antarmuka dashboard interaktif yang menampilkan kartu indikator, filter dinamis, tabel kepuasan, grafik ApexCharts, modal, dan fitur drill-down.

```

1  @extends('layouts.app')
2
3  @section('title', 'Dashboard')
4
5  @section('content')
6  <style>
7      :root {
8          --brand-50: #fde8ec;
9          --brand-100: #fbbcca;
10         --brand-500: #8b1a2a;
11         --brand-700: #6c0215;
12         --slate-50: #f8fafc;
13         --slate-100: #f1f5f9;
14         --slate-200: #e2e8f0;
15         --slate-400: #94a3b8;
16         --slate-500: #64748b;
17         --slate-700: #334155;
18         --slate-900: #0f172a;
19         --radius: 12px;
20         --shadow: 0 1px 3px rgba(0,0,0,.06), 0 1px 2px rgba(0,0,0,.04);
21         --shadow-md: 0 4px 16px rgba(0,0,0,.07);
22     }
23
24     .db-wrap { display: flex; flex-direction: column; gap: 1rem; }
25     .db-header { display: flex; justify-content: space-between; align-items: flex-end; flex-wrap: wrap; gap: .75rem; }
26     .db-header h3 { font-size: 1.25rem; font-weight: 700; color: var(--slate-900); margin: 0; }
27     .db-header p { font-size: .82rem; color: var(--slate-400); margin: .2rem 0 0; }
28
29     .filter-bar {
30         background: #fff; border: 1px solid var(--slate-200);
31         border-radius: var(--radius); padding: .75rem 1rem; box-shadow: var(--shadow);
32     }
33     .filter-bar .form-label {
34         font-size: .7rem; font-weight: 700; color: var(--slate-500);
35         text-transform: uppercase; letter-spacing: .5px; margin-bottom: .25rem;
36     }
37     .filter-bar .form-select,
38     .filter-bar .periode-toggle {
39         font-size: .82rem; border-color: var(--slate-200); border-radius: 8px; color: var(--slate-700);
40     }
41     .filter-bar .form-select:focus,
42     .filter-bar .periode-toggle:focus { border-color: var(--brand-500); box-shadow: 0 0 3px rgba(139,26,42,.12); }
43     .periode-dropdown { width: 100%; }
44     .periode-toggle {
45         width: 100%; min-height: 31px; background: #fff; border: 1px solid var(--slate-200);
46         display: flex; align-items: center; justify-content: space-between; gap: .5rem;

```

```

47     padding: .25rem .5rem; text-align: left;
48 }
49 .periode-toggle span { overflow: hidden; text-overflow: ellipsis; white-space: nowrap; }
50 .periode-menu {
51     width: 100%; max-height: 220px; overflow-y: auto; padding: .45rem;
52     border: 1px solid var(--slate-200); border-radius: 8px; box-shadow: var(--shadow-md);
53 }
54 .periode-option {
55     display: flex; align-items: center; gap: .45rem; padding: .32rem .4rem;
56     border-radius: 6px; font-size: .82rem; color: var(--slate-700); cursor: pointer;
57 }
58 .periode-option:hover { background: var(--slate-50); }
59 .periode-option input { accent-color: var(--brand-500); }
60 .btn-apply {
61     font-size: .8rem; font-weight: 600; padding: .42rem 1rem;
62     background: var(--brand-500); color: #fff; border: none; border-radius: 8px;
63 }
64 .btn-apply:hover { background: var(--brand-700); color: #fff; }
65 .btn-reset { font-size: .8rem; padding: .42rem .85rem; border-radius: 8px; }
66 .btn-extend {
67     border: 1px solid var(--slate-200); background: #fff; color: var(--slate-700);
68     border-radius: 8px; font-size: .75rem; font-weight: 600; padding: .32rem .65rem;
69     display: inline-flex; align-items: center; gap: .3rem;
70 }
71 .btn-extend:hover { border-color: var(--brand-100); color: var(--brand-700); background: var(--brand-50); }
72 .panel-actions { display: inline-flex; align-items: center; gap: .4rem; flex-wrap: wrap; }
73 .btn-back {
74     display: none; border: 1px solid var(--brand-100); background: var(--brand-50); color: var(--brand-700);
75     border-radius: 8px; font-size: .75rem; font-weight: 700; padding: .32rem .65rem;
76     align-items: center; gap: .3rem;
77 }
78 .btn-back.show { display: inline-flex; }
79 .chip {
80     display: inline-flex; align-items: center; gap: .3rem;
81     font-size: .72rem; font-weight: 500; padding: .2rem .6rem;
82     border-radius: 50px; background: var(--brand-50);
83     color: var(--brand-700); border: 1px solid var(--brand-100);
84 }
85
86 .stat-grid { display: grid; grid-template-columns: repeat(4, 1fr); gap: .85rem; }
87 @media(max-width:1199px) { .stat-grid { grid-template-columns: repeat(2, 1fr); } }
88 @media(max-width:575px) { .stat-grid { grid-template-columns: 1fr; gap: .7rem; } }
89
90 .panel, .stat-card {
91     background: #fff; border: 1px solid var(--slate-200);
92     border-radius: var(--radius); box-shadow: var(--shadow); overflow: hidden;
93 }
94 .panel:hover, .stat-card:hover { box-shadow: var(--shadow-md); }
95 .stat-card {
96     padding: 1rem 1.1rem; position: relative; transition: box-shadow .2s, transform .2s;
97 }
98 .stat-card:hover { transform: translateY(-2px); }
99 .stat-card::before {

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100     content: ''; position: absolute; top: 0; left: 0; right: 0; height: 3px; background: var(--brand-500);
101   }
102   .stat-card.green::before { background: #16a34a; }
103   .stat-card.amber::before { background: #d97706; }
104   .stat-card.red::before { background: #dc2626; }
105   .stat-top { display: flex; justify-content: space-between; align-items: flex-start; gap: .75rem; margin-bottom: .7rem; }
106   .stat-label {
107     font-size: .7rem; font-weight: 700; text-transform: uppercase;
108     letter-spacing: .55px; color: var(--slate-500);
109   }
110   .stat-icon-wrap {
111     width: 34px; height: 34px; border-radius: 9px;
112     display: flex; align-items: center; justify-content: center;
113     font-size: .95rem; background: var(--brand-50); color: var(--brand-500); flex-shrink: 0;
114   }
115   .stat-card.green .stat-icon-wrap { background: #f0fdf4; color: #16a34a; }
116   .stat-card.amber .stat-icon-wrap { background: #fffbeb; color: #d97706; }
117   .stat-card.red .stat-icon-wrap { background: #fef2f2; color: #dc2626; }
118   .stat-value { font-size: 1.65rem; font-weight: 800; color: var(--slate-900); line-height: 1; }
119   .stat-unit { font-size: .76rem; font-weight: 400; color: var(--slate-500); margin-left: .25rem; }
120   .stat-name { font-size: .88rem; font-weight: 700; color: var(--slate-900); line-height: 1.25; margin-bottom: .22rem; }
121   .stat-sub { font-size: .72rem; color: var(--slate-500); margin-top: .35rem; line-height: 1.35; }
122   .stat-bar-track { height: 4px; background: var(--slate-100); border-radius: 99px; margin-top: .55rem; overflow: hidden; }
123   .stat-bar-fill { height: 100%; border-radius: 99px; background: #16a34a; }
124
125   .chart-grid { display: grid; grid-template-columns: repeat(2, minmax(0, 1fr)); gap: .85rem; align-items: start; }
126   @media(max-width:991px) { .chart-grid { grid-template-columns: 1fr; } }
127
128   .panel-header {
129     display: flex; justify-content: space-between; align-items: center;
130     padding: .75rem 1rem; border-bottom: 1px solid var(--slate-100); background: #fff;
131     gap: .75rem; flex-wrap: wrap;
132   }
133   .panel-title { font-size: .86rem; font-weight: 700; color: var(--slate-700); margin: 0; }
134   .panel-subtitle { font-size: .72rem; color: var(--slate-500); margin: .08rem 0 0; }
135   .panel-body { padding: .75rem 1rem .9rem; }
136   .chart-wrap { min-height: 0; }
137   .chart-compact { height: 240px; }
138   .chart-modal { min-height: 420px; }
139
140   .feedback-list-main .fb-item:nth-child(n+4) { display: none; }
141   .fb-item { padding: .75rem 1rem; border-bottom: 1px solid var(--slate-100); }
142   .fb-item:last-child { border-bottom: none; }
143   .fb-quote {
144     font-size: .8rem; color: var(--slate-700); line-height: 1.5;
145     font-style: italic; padding: .5rem .75rem;
146     background: var(--brand-50); border-left: 3px solid var(--brand-500);
147     border-radius: 0 8px 8px 0; margin-bottom: .45rem;
148   }
149   .fb-meta { display: flex; justify-content: space-between; align-items: center; flex-wrap: wrap; gap: .35rem; }
150   .fb-who { font-size: .74rem; }
151   .fb-who strong { color: var(--slate-700); font-weight: 600; }
152   .fb-who span { color: var(--slate-500); }

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153 .fb-tag {
154     font-size: .68rem; font-weight: 500; padding: .16rem .55rem;
155     background: var(--brand-50); color: var(--brand-700);
156     border: 1px solid var(--brand-100); border-radius: 99px;
157 }
158 .feedback-modal-body { max-height: 70vh; overflow-y: auto; padding: 0; }
159
160 .drill-card { padding: 1rem; display: grid; grid-template-columns: 220px 1fr; gap: 1rem; align-items: center; }
161 @media(max-width:767px) { .drill-card { grid-template-columns: 1fr; } }
162 .drill-score {
163     min-height: 150px; border: 1px solid var(--slate-200); border-radius: 10px;
164     display: flex; flex-direction: column; align-items: center; justify-content: center;
165     background: var(--slate-50); text-align: center; padding: 1rem;
166 }
167 .drill-score strong { font-size: 2rem; font-weight: 800; color: var(--slate-900); line-height: 1; }
168 .drill-score span { font-size: .74rem; color: var(--slate-500); margin-top: .35rem; }
169 .drill-title { font-size: .98rem; font-weight: 800; color: var(--slate-900); margin-bottom: .2rem; }
170 .drill-subtitle { font-size: .74rem; color: var(--slate-500); margin-bottom: .85rem; }
171 .rating-row { display: grid; grid-template-columns: 130px 1fr 62px; gap: .65rem; align-items: center; margin-bottom: .58rem; }
172 .rating-label { font-size: .76rem; color: var(--slate-700); font-weight: 600; }
173 .rating-track { height: 9px; background: var(--slate-100); border-radius: 99px; overflow: hidden; }
174 .rating-fill { height: 100%; border-radius: 99px; width: 0; transition: width .22s ease; }
175 .rating-fill.sb { background: #16a34a; }
176 .rating-fill.b { background: #3b82f6; }
177 .rating-fill.k { background: #d97706; }
178 .rating-fill.sk { background: #dc2626; }
179 .rating-value { font-size: .74rem; color: var(--slate-500); text-align: right; }
180 .view-switch { display: inline-flex; border: 1px solid var(--slate-200); border-radius: 8px; overflow: hidden; }
181 .view-switch button {
182     border: 0; background: #fff; color: var(--slate-500);
183     padding: .32rem .55rem; font-size: .74rem; font-weight: 700;
184 }
185 .view-switch button.active { background: var(--brand-500); color: #fff; }
186 .kepuasan-chart-wrap { display: none; min-height: 280px; padding: .75rem 1rem 1rem; }
187 .satisfaction-panel.chart-mode .kepuasan-table-wrap { display: none; }
188 .satisfaction-panel.chart-mode .kepuasan-chart-wrap { display: block; }
189 .tbl-kepuasan tbody tr { cursor: pointer; }
190 .tbl-kepuasan tbody tr.active td { background: var(--brand-50); color: var(--brand-700) !important; }
191
192 .satisfaction-panel, .satisfaction-panel * { color: #111 !important; }
193 .satisfaction-panel .view-switch button { color: var(--slate-500) !important; }
194 .satisfaction-panel .view-switch button.active { color: #fff !important; background: var(--brand-500); }
195 .satisfaction-panel .panel-header { border: 1px solid #d1d5db; border-width: 0 0 1px 0; background: #fff; }
196 .tbl-kepuasan { width: 100%; border-collapse: collapse; font-size: .81rem; }
197 .tbl-kepuasan thead th {
198     padding: .56rem .9rem; font-size: .7rem; font-weight: 700;
199     text-align: center; white-space: nowrap; letter-spacing: .3px;
200     border-bottom: 1px solid #111; background: #fff;
201 }
202 .tbl-kepuasan thead .th-name { text-align: left; width: 26%; }
203 .tbl-kepuasan tbody td {
204     padding: .52rem .9rem; border-bottom: 1px solid var(--slate-100);
205     text-align: center; vertical-align: middle;

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206     }
207     .tbl-kepuasan tbody td:first-child { text-align: left; font-weight: 500; }
208     .tbl-kepuasan tbody tr:nth-child(even) td { background: var(--slate-50); }
209     .tbl-kepuasan tbody tr:hover td { background: #f3f4f6; transition: background .1s; }
210     .tbl-kepuasan tfoot td {
211         padding: .56rem .9rem; font-weight: 700; text-align: center;
212         font-size: .8rem; border-top: 1px solid #111;
213     }
214     .tbl-kepuasan tfoot td:first-child { text-align: left; }
215     .empty-state { text-align: center; padding: 1.75rem 1rem; color: var(--slate-500); font-size: .84rem; }
216     .empty-state i { display: block; font-size: 1.5rem; margin-bottom: .45rem; opacity: .45; }
217
218     .modal-content { border: 0; border-radius: var(--radius); overflow: hidden; }
219     .modal-header { border-bottom-color: var(--slate-100); padding: .85rem 1rem; }
220     .modal-title { font-size: .95rem; font-weight: 700; color: var(--slate-900); }
221 </style>
222
223 @php
224     $activePeriode = $filters['periode'] ?? [];
225     $selectedFakultas = $filters['fakultas'] ?? '';
226     $selectedProdi = $filters['program_studi'] ?? '';
227     $fakultasProdi = $filterOptions['fakultasProdi'] ?? [];
228     $fakultasLabels = $filterOptions['fakultasLabels'] ?? [];
229     $availableProdi = $selectedFakultas && isset($fakultasProdi[$selectedFakultas])
230         ? $fakultasProdi[$selectedFakultas]
231         : $filterOptions['prodiList'];
232     $hasFilters = !empty($activePeriode) || !empty($filters['fakultas']) || !empty($filters['program_studi']);
233     $pct = min(100, round((( $rataKeseluruhan ?? 0) / 4) * 100));
234     $activeKategori = $kategoriTerlemah->kategori ?? $kategoriTerbaik->kategori ?? null;
235 @endphp
236
237 <div class="db-wrap">
238     <div class="db-header">
239         <div>
240             <h3>Dashboard Evaluasi Lulusan</h3>
241             <p>Ringkasan performa dan kualitas lulusan Universitas Dinamika di dunia kerja.</p>
242         </div>
243     </div>
244
245     <div class="filter-bar">
246         <form method="GET" action="{ route('dashboard') }" id="filterForm">
247             <div class="row g-2 align-items-end">
248                 <div class="col-12 col-lg-auto d-flex align-items-center pe-3">
249                     <span style="font-size:.72rem;font-weight:700;text-transform:uppercase;letter-spacing:.6px;color:var(--slate-500);">
250                         <i class="bi bi-sliders me-1" style="color:var(--brand-500);"></i>Filter
251                     </span>
252                 </div>
253                 <div class="col-12 col-md-4 col-lg">
254                     <label class="form-label">Periode</label>
255                     <div class="dropdown periode-dropdown">
256                         <button class="periode-toggle" type="button" data-bs-toggle="dropdown" data-bs-auto-close="outside" aria-expanded="false">
257                             <span>
258                                 @if(empty($activePeriode))

```

```

259         Semua Periode
260         @elseif(count($activePeriode) === 1)
261             {{ $activePeriode[0] }}
262         @else
263             {{ count($activePeriode) }} periode dipilih
264         @endif
265     </span>
266     <i class="bi bi-chevron-down"></i>
267 </button>
268 <div class="dropdown-menu periode-menu">
269     @foreach($filterOptions['periodeList'] as $periode)
270         <label class="periode-option">
271             <input type="checkbox" name="periode[]" value="{{ $periode }}" {{ in_array($periode, $activePeriode, true) ? 'checked' : '' }}>
272             <span>{{ $periode }}</span>
273         </label>
274     @endforeach
275 </div>
276 </div>
277 </div>
278 <div class="col-12 col-md-4 col-lg">
279     <label class="form-label">Fakultas</label>
280     <select name="fakultas" id="filterFakultas" class="form-select form-select-sm">
281         <option value="">Semua Fakultas</option>
282         @foreach($filterOptions['fakultasList'] as $fakultas)
283             <option value="{{ $fakultas }}" {{ $selectedFakultas == $fakultas ? 'selected' : '' }}>
284                 {{ $fakultasLabels[$fakultas] ?? $fakultas }}
285             </option>
286         @endforeach
287     </select>
288 </div>
289 <div class="col-12 col-md-4 col-lg">
290     <label class="form-label">Program Studi</label>
291     <select name="program_studi" id="filterProdi" class="form-select form-select-sm">
292         <option value="">{{ $selectedFakultas ? 'Semua Prodi di Fakultas ini' : 'Semua Prodi' }}</option>
293         @foreach($availableProdi as $prodi)
294             <option value="{{ $prodi }}" {{ $selectedProdi == $prodi ? 'selected' : '' }}>{{ $prodi }}</option>
295         @endforeach
296     </select>
297 </div>
298 <div class="col-12 col-lg-auto d-flex gap-2">
299     <button type="submit" class="btn btn-apply">
300         <i class="bi bi-search me-1"></i> Terapkan
301     </button>
302     @if($hasFilters)
303         <a href="{{ route('dashboard') }}" class="btn btn-reset btn-outline-secondary">
304             <i class="bi bi-x-lg"></i>
305         </a>
306     @endif
307 </div>
308 </div>
309
310 @if($hasFilters)
311     <div class="d-flex flex-wrap gap-2 align-items-center mt-2 pt-2 border-top">

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312         <span style="font-size:.72rem;color:var(--slate-500);">Aktif:</span>
313         @foreach($activePeriode as $periode)
314             <span class="chip"><i class="bi bi-calendar3"></i> {{ $periode }}</span>
315         @endforeach
316         @if(!empty($filters['fakultas']))
317             <span class="chip"><i class="bi bi-building"></i> {{ $fakultasLabels[$filters['fakultas']] ?? $filters['fakultas'] }}</span>
318         @endif
319         @if(!empty($filters['program_studi']))
320             <span class="chip"><i class="bi bi-book"></i> {{ $filters['program_studi'] }}</span>
321         @endif
322     </div>
323 @endif
324 </form>
325 </div>
326
327 <div class="stat-grid">
328     <div class="stat-card">
329         <div class="stat-top">
330             <span class="stat-label">Lulusan Dinilai</span>
331             <div class="stat-icon-wrap"><i class="bi bi-mortarboard-fill"></i></div>
332         </div>
333         <span class="stat-value">{{ $totalLulusan ?? 0 }}</span>
334         <span class="stat-unit">orang</span>
335         <div class="stat-sub">{{ $totalSurvey ?? 0 }} respon survey terarsip</div>
336     </div>
337
338     <div class="stat-card green">
339         <div class="stat-top">
340             <span class="stat-label">Indeks Kepuasan</span>
341             <div class="stat-icon-wrap"><i class="bi bi-star-fill"></i></div>
342         </div>
343         <span class="stat-value">{{ number_format($rataKeseluruhan ?? 0, 2) }}</span>
344         <span class="stat-unit">/ 4.00</span>
345         <div class="stat-bar-track">
346             <div class="stat-bar-fill" style="width:{{ $pct }}%;"></div>
347         </div>
348         <div class="stat-sub">{{ $pct }}% dari skor maksimal</div>
349     </div>
350
351     <div class="stat-card amber">
352         <div class="stat-top">
353             <span class="stat-label">Kategori Terbaik</span>
354             <div class="stat-icon-wrap"><i class="bi bi-trophy-fill"></i></div>
355         </div>
356         <div class="stat-name">{{ $kategoriTerbaik->kategori ?? '-' }}</div>
357         <div class="stat-sub">Skor {{ number_format($kategoriTerbaik->rata_rata ?? 0, 2) }} / 4.00</div>
358     </div>
359
360     <div class="stat-card red">
361         <div class="stat-top">
362             <span class="stat-label">Kategori Terendah</span>
363             <div class="stat-icon-wrap"><i class="bi bi-arrow-down-right"></i></div>
364         </div>

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365     <div class="stat-name">{{ $kategoriTerlemah->kategori ?? '-' }}</div>
366     <div class="stat-sub">Skor {{ number_format($kategoriTerlemah->rata_rata ?? 0, 2) }} / 4.00</div>
367   </div>
368 </div>
369
370 <div class="chart-grid">
371   <div class="panel">
372     <div class="panel-header">
373       <div>
374         <h6 class="panel-title" id="prodiPanelTitle">Responden Berdasarkan Program Studi</h6>
375         <p class="panel-subtitle" id="prodiPanelSubtitle">Klik bar untuk melihat detail jenis perusahaan pada prodi tersebut</p>
376       </div>
377       <div class="panel-actions">
378         <button type="button" class="btn-back" id="backProdiChart">
379           <i class="bi bi-arrow-left"></i> Ringkasan
380         </button>
381         <button type="button" class="btn-extend" data-bs-toggle="modal" data-bs-target="#prodiChartModal">
382           <i class="bi bi-arrows-fullscreen"></i> Lihat Selengkapnya
383         </button>
384       </div>
385     </div>
386     <div class="panel-body">
387       <div id="chart-prodi" class="chart-wrap chart-compact"></div>
388     </div>
389   </div>
390
391   <div class="panel">
392     <div class="panel-header">
393       <div>
394         <h6 class="panel-title" id="kinerjaPanelTitle">Rata-Rata Penilaian per Kategori</h6>
395         <p class="panel-subtitle" id="kinerjaPanelSubtitle">Klik bar untuk melihat distribusi jawaban kategori tersebut</p>
396       </div>
397       <div class="panel-actions">
398         <button type="button" class="btn-back" id="backKinerjaChart">
399           <i class="bi bi-arrow-left"></i> Ringkasan
400         </button>
401         <button type="button" class="btn-extend" data-bs-toggle="modal" data-bs-target="#kinerjaChartModal">
402           <i class="bi bi-arrows-fullscreen"></i> Lihat Selengkapnya
403         </button>
404       </div>
405     </div>
406     <div class="panel-body">
407       <div id="chart-kinerja" class="chart-wrap chart-compact"></div>
408     </div>
409   </div>
410 </div>
411
412 <div class="panel">
413   <div class="panel-header">
414     <div>
415       <h6 class="panel-title">Umpan Balik Terbaru</h6>
416       <p class="panel-subtitle">Tiga feedback terbaru ditampilkan di dashboard utama</p>
417     </div>

```

```

418     <button type="button" class="btn-extend" data-bs-toggle="modal" data-bs-target="#feedbackModal">
419         <i class="bi bi-chat-square-text"></i> Lihat Semua
420     </button>
421 </div>
422 <div class="feedback-list-main">
423     @foreach($komentarTerbaru->take(3) as $komen)
424         @include('admin.dashboard.partials.feedback-item', ['komen' => $komen])
425     @empty
426         <div class="empty-state"><i class="bi bi-chat-dots"></i>Belum ada umpan balik.</div>
427     @endforeach
428 </div>
429 </div>
430
431 <div class="panel satisfaction-panel">
432     <div class="panel-header">
433         <div>
434             <h6 class="panel-title">Tingkat Kepuasan Pengguna</h6>
435             <p class="panel-subtitle">Persentase penilaian responden per kategori kompetensi lulusan</p>
436         </div>
437         @if(!$kepuasanPerKategori->isEmpty())
438             <div class="view-switch" role="group" aria-label="Mode tampilan kepuasan">
439                 <button type="button" class="active" data-view-mode="table"><i class="bi bi-table"></i></button>
440                 <button type="button" data-view-mode="chart"><i class="bi bi-bar-chart"></i></button>
441             </div>
442         @endif
443     </div>
444
445     @if($kepuasanPerKategori->isEmpty())
446         <div class="empty-state"><i class="bi bi-bar-chart"></i>Belum ada data penilaian.</div>
447     @else
448         <div class="kepuasan-table-wrap" style="overflow-x:auto;">
449             <table class="tbl-kepuasan">
450                 <thead>
451                     <tr>
452                         <th class="th-name">Jenis Kemampuan</th>
453                         <th>Sangat Baik (4)</th>
454                         <th>Baik (3)</th>
455                         <th>Kurang (2)</th>
456                         <th>Sangat Kurang (1)</th>
457                     </tr>
458                 </thead>
459                 <tbody>
460                     @foreach($kepuasanPerKategori as $kat)
461                         <tr data-kategori-row="{ $kat['kategori'] }">
462                             <td>{{ $kat['kategori'] }}</td>
463                             <td>{{ $kat['pct_sb'] }}%</td>
464                             <td>{{ $kat['pct_b'] }}%</td>
465                             <td>{{ $kat['pct_k'] }}%</td>
466                             <td>{{ $kat['pct_sk'] }}%</td>
467                         </tr>
468                     @endforeach
469                 </tbody>
470             </table>

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```

471         <tr>
472             <td>Total</td>
473             <td>{{ $kepuasanRingkasan['total']['sb'] }}%</td>
474             <td>{{ $kepuasanRingkasan['total']['b'] }}%</td>
475             <td>{{ $kepuasanRingkasan['total']['k'] }}%</td>
476             <td>{{ $kepuasanRingkasan['total']['sk'] }}%</td>
477         </tr>
478         <tr>
479             <td>Rata-Rata</td>
480             <td>{{ $kepuasanRingkasan['rata']['sb'] }}%</td>
481             <td>{{ $kepuasanRingkasan['rata']['b'] }}%</td>
482             <td>{{ $kepuasanRingkasan['rata']['k'] }}%</td>
483             <td>{{ $kepuasanRingkasan['rata']['sk'] }}%</td>
484         </tr>
485     </tfoot>
486 </table>
487 </div>
488 <div class="kepuasan-chart-wrap">
489     <div id="chart-kepuasan-stack"></div>
490 </div>
491 @endif
492 </div>
493 </div>
494
495 <div class="modal fade" id="prodiChartModal" tabindex="-1" aria-labelledby="prodiChartModallabel" aria-hidden="true">
496     <div class="modal-dialog modal-xl modal-dialog-centered modal-dialog-scrollable">
497         <div class="modal-content">
498             <div class="modal-header">
499                 <h5 class="modal-title" id="prodiChartModallabel">Responden Berdasarkan Program Studi</h5>
500                 <div class="panel-actions ms-auto">
501                     <button type="button" class="btn-back" id="backProdiFullChart">
502                         <i class="bi bi-arrow-left"></i> Ringkasan
503                     </button>
504                 </div>
505                 <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
506             </div>
507             <div class="modal-body">
508                 <div id="chart-prodi-full" class="chart-modal"></div>
509             </div>
510         </div>
511     </div>
512 </div>
513
514 <div class="modal fade" id="kinerjaChartModal" tabindex="-1" aria-labelledby="kinerjaChartModallabel" aria-hidden="true">
515     <div class="modal-dialog modal-xl modal-dialog-centered modal-dialog-scrollable">
516         <div class="modal-content">
517             <div class="modal-header">
518                 <h5 class="modal-title" id="kinerjaChartModallabel">Rata-Rata Penilaian per Kategori</h5>
519                 <div class="panel-actions ms-auto">
520                     <button type="button" class="btn-back" id="backKinerjaFullChart">
521                         <i class="bi bi-arrow-left"></i> Ringkasan
522                     </button>
523                 </div>

```

```

524         <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
525     </div>
526     <div class="modal-body">
527         <div id="chart-kinerja-full" class="chart-modal"></div>
528     </div>
529 </div>
530 </div>
531 </div>
532
533 <div class="modal fade" id="feedbackModal" tabindex="-1" aria-labelledby="feedbackModalLabel" aria-hidden="true">
534     <div class="modal-dialog modal-lg modal-dialog-centered modal-dialog-scrollable">
535         <div class="modal-content">
536             <div class="modal-header">
537                 <h5 class="modal-title" id="feedbackModalLabel">Seluruh Umpan Balik</h5>
538                 <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
539             </div>
540             <div class="modal-body feedback-modal-body">
541                 @forelse($komentarTerbaru as $komen)
542                     @include('admin.dashboard.partials.feedback-item', ['komen' => $komen])
543                 @empty
544                     <div class="empty-state"><i class="bi bi-chat-dots"></i>Belum ada umpan balik.</div>
545                 @endforelse
546             </div>
547         </div>
548     </div>
549 </div>
550
551 <script src="{ asset('assets/vendors/apexcharts/apexcharts.min.js') }"></script>
552 <script>
553     const chartData = @json($chartData);
554     const chartLabels = @json($chartLabels);
555     const respondenProdiData = @json($respondenProdiData);
556     const respondenProdiLabels = @json($respondenProdiLabels);
557     const prodiDetails = @json($prodiDetails);
558     const fakultasProdi = @json($fakultasProdi);
559     const kategoriDetails = @json($kategoriDetails);
560     const initialKategori = @json($activeKategori);
561     const compactLimit = 6;
562     const kategoriDetailMap = new Map(kategoriDetails.map((item) => [item.kategori, item]));
563     const prodiDetailMap = new Map(prodiDetails.map((item) => [item.prodi, item]));
564
565     const fakultasSelect = document.getElementById('filterFakultas');
566     const prodiSelect = document.getElementById('filterProdi');
567
568     const renderProdiOptions = () => {
569         const selectedFakultas = fakultasSelect.value;
570         const currentProdi = prodiSelect.value;
571         const prodiList = selectedFakultas ? (fakultasProdi[selectedFakultas] || []) : Object.values(fakultasProdi).flat();
572         const defaultLabel = selectedFakultas ? 'Semua Prodi di Fakultas ini' : 'Semua Prodi';
573
574         prodiSelect.disabled = true;
575         prodiSelect.innerHTML = '<option value="">Memuat prodi...</option>';
576

```

```

577     window.setTimeout(() => {
578         prodiSelect.innerHTML = '';
579         prodiSelect.append(new Option(defaultLabel, ''));
580
581         prodiList.forEach((prodi) => {
582             prodiSelect.append(new Option(prodi, prodi));
583         });
584
585         prodiSelect.value = prodiList.includes(currentProdi) ? currentProdi : '';
586         prodiSelect.disabled = false;
587     }, 120);
588 };
589
590 fakultasSelect.addEventListener('change', renderProdiOptions);
591
592 const sliceData = (labels, data, limit = compactLimit) => ({
593     labels: labels.slice(0, limit),
594     data: data.slice(0, limit)
595 });
596
597 const emptyChart = (selector, message) => {
598     const target = document.querySelector(selector);
599     if (target) target.innerHTML = `<div class="empty-state"><i class="bi bi-bar-chart"></i>${message}</div>`;
600 };
601
602 const chartHeight = (count, min = 220, row = 34, max = 420) => Math.min(max, Math.max(min, count * row + 46));
603 const formatPct = (value) => `${Number(value || 0).toFixed(1)}%`;
604 const ratingLabels = {
605     sb: 'Sangat Baik',
606     b: 'Baik',
607     k: 'Kurang',
608     sk: 'Sangat Kurang'
609 };
610
611 const setActiveKategori = (kategori) => {
612     const detail = kategoriDetailMap.get(kategori) || kategoriDetails[0];
613     if (!detail) return;
614
615     document.querySelectorAll('[data-kategori-row]').forEach((row) => {
616         row.classList.toggle('active', row.dataset.kategoriRow === detail.kategori);
617     });
618 };
619
620 const setPanelText = (titleId, subtitleId, title, subtitle) => {
621     const titleEl = document.getElementById(titleId);
622     const subtitleEl = document.getElementById(subtitleId);
623
624     if (titleEl) titleEl.textContent = title;
625     if (subtitleEl) subtitleEl.textContent = subtitle;
626 };
627
628 const setBackVisible = (buttonId, visible) => {
629     document.getElementById(buttonId)?.classList.toggle('show', visible);

```



```

630   };
631
632   const prodiOptions = (labels, data, expanded = false, onSelect = null) => ({
633     series: [{ name: 'Responden', data }],
634     chart: {
635       type: 'bar',
636       height: expanded ? chartHeight(labels.length, 320, 38, 760) : 240,
637       toolbar: { show: false },
638       fontFamily: 'inherit',
639       events: onSelect ? {
640         dataPointSelection: (event, chartContext, config) => {
641           const prodi = labels[config.dataPointIndex];
642           if (prodi) onSelect(prodi);
643         }
644       } : {}
645     },
646     plotOptions: {
647       bar: {
648         horizontal: true,
649         borderRadius: 5,
650         barHeight: expanded ? '54%' : '48%',
651         dataLabels: { position: 'center' }
652       }
653     },
654     colors: ['#8b1a2a'],
655     dataLabels: {
656       enabled: true,
657       formatter: value => `${value}`,
658       style: { colors: ['fff'], fontSize: '11px', fontWeight: 700 }
659     },
660     xaxis: {
661       categories: labels,
662       decimalsInFloat: 0,
663       labels: {
664         trim: true,
665         style: { colors: '#64748b', fontSize: '11px' }
666       },
667       axisBorder: { show: false },
668       axisTicks: { show: false }
669     },
670     yaxis: { labels: { style: { colors: '#334155', fontSize: '11px', fontWeight: 500 } } },
671     grid: { borderColor: '#f1f5f9', strokeDashArray: 4 },
672     tooltip: { y: { title: { formatter: () => 'Responden:' } } }
673   });
674
675   const prodiDetailOptions = (detail, expanded = false) => {
676     const items = detail.jenis_perusahaan || [];
677     const labels = items.map((item) => item.label);
678     const data = items.map((item) => item.total);
679
680     return {
681       series: [{ name: 'Responden', data }],
682       chart: { type: 'bar', height: expanded ? 420 : 240, toolbar: { show: false }, fontFamily: 'inherit' },

```

```

683     plotOptions: { bar: { horizontal: false, borderRadius: 5, columnWidth: expanded ? '36%' : '45%', dataLabels: { position: 'top' } } },
684     colors: ['#3b82f6'],
685     dataLabels: {
686       enabled: true,
687       offsetY: -18,
688       style: { colors: ['#334155'], fontSize: '11px', fontWeight: 700 },
689       formatter: value => `${value}`
690     },
691     xaxis: {
692       categories: labels,
693       labels: { trim: true, rotate: expanded ? -30 : -20, hideOverlappingLabels: true, style: { colors: '#64748b', fontSize: expanded ? '11px' : '10px' } },
694       axisBorder: { show: false },
695       axisTicks: { show: false }
696     },
697     yaxis: {
698       min: 0,
699       decimalsInFloat: 0,
700       labels: { formatter: value => Number(value).toFixed(0), style: { colors: '#334155', fontSize: '11px', fontWeight: 500 } }
701     },
702     grid: { borderColor: '#f1f5f9', strokeDashArray: 4 },
703     tooltip: { y: { title: { formatter: () => 'Responden:' } } }
704   };
705 };
706
707 const kinerjaOptions = (labels, data, expanded = false, onSelect = null) => ({
708   series: [{ name: 'Skor', data }],
709   chart: {
710     type: 'bar',
711     height: expanded ? 420 : 240,
712     toolbar: { show: false },
713     fontFamily: 'inherit',
714     events: onSelect ? {
715       dataPointSelection: (event, chartContext, config) => {
716         const kategori = labels[config.dataPointIndex];
717         if (kategori) onSelect(kategori);
718       }
719     } : {}
720   },
721   plotOptions: {
722     bar: {
723       horizontal: false,
724       borderRadius: 5,
725       columnWidth: expanded ? '44%' : '48%',
726       dataLabels: { position: 'top' }
727     }
728   },
729   colors: ['#8b1a2a'],
730   dataLabels: {
731     enabled: true,
732     offsetY: -18,
733     style: { colors: ['#334155'], fontSize: '11px', fontWeight: 700 },
734     formatter: value => Number(value).toFixed(2)
735   },

```

```

736     xaxis: {
737       categories: labels,
738       labels: {
739         trim: true,
740         rotate: expanded ? -35 : -25,
741         hideOverlappingLabels: true,
742         style: { colors: '#64748b', fontSize: expanded ? '11px' : '10px' }
743       },
744       axisBorder: { show: false },
745       axisTicks: { show: false }
746     },
747     yaxis: {
748       min: 0,
749       max: 4,
750       tickAmount: 4,
751       labels: {
752         formatter: value => Number(value).toFixed(0),
753         style: { colors: '#334155', fontSize: '11px', fontWeight: 500 }
754       }
755     },
756     grid: { yaxis: { lines: { show: true } }, borderColor: '#f1f5f9', strokeDashArray: 4 },
757     tooltip: { y: { title: { formatter: () => 'Skor:' } } }
758   });
759
760   const kategoriDetailOptions = (detail, expanded = false) => {
761     const keys = ['sb', 'b', 'k', 'sk'];
762
763     return {
764       series: [{ name: 'Persentase', data: keys.map((key) => detail.percentages?.[key] || 0) }],
765       chart: { type: 'bar', height: expanded ? 420 : 240, toolbar: { show: false }, fontFamily: 'inherit' },
766       plotOptions: { bar: { horizontal: true, borderRadius: 5, barHeight: expanded ? '46%' : '52%', dataLabels: { position: 'center' } } },
767       colors: ['#16a34a'],
768       dataLabels: {
769         enabled: true,
770         formatter: (value, opts) => {
771           const key = keys[opts.dataPointIndex];
772           return `${formatPct(value)} (${detail.counts?.[key] || 0})`;
773         },
774         style: { colors: ['fff'], fontSize: '11px', fontWeight: 700 }
775       },
776       xaxis: {
777         min: 0,
778         max: 100,
779         categories: keys.map((key) => ratingLabels[key]),
780         labels: { formatter: (value) => `${Number(value).toFixed(0)}%`, style: { colors: '#64748b', fontSize: '11px' } },
781         axisBorder: { show: false },
782         axisTicks: { show: false }
783       },
784       yaxis: { labels: { style: { colors: '#334155', fontSize: '11px', fontWeight: 600 } } },
785       grid: { borderColor: '#f1f5f9', strokeDashArray: 4 },
786       tooltip: { y: { formatter: (value) => formatPct(value), title: { formatter: () => 'Porsi:' } } }
787     };
788   };

```

```

789
790 const kepuasanStackOptions = () => ({
791   series: [
792     { name: 'Sangat Baik', data: kategoriDetails.map((item) => item.percentages.sb) },
793     { name: 'Baik', data: kategoriDetails.map((item) => item.percentages.b) },
794     { name: 'Kurang', data: kategoriDetails.map((item) => item.percentages.k) },
795     { name: 'Sangat Kurang', data: kategoriDetails.map((item) => item.percentages.sk) },
796   ],
797   chart: { type: 'bar', height: 300, stacked: true, stackType: '100%', toolbar: { show: false }, fontFamily: 'inherit' },
798   plotOptions: { bar: { horizontal: true, borderRadius: 4, barHeight: '58%' } },
799   colors: ['#16a34a', '#3b82f6', '#d97706', '#dc2626'],
800   xaxis: {
801     categories: kategoriDetails.map((item) => item.kategori),
802     labels: { formatter: (value) => `${Number(value).toFixed(0)}%`, style: { colors: '#64748b', fontSize: '11px' } },
803   },
804   yaxis: { labels: { style: { colors: '#334155', fontSize: '11px', fontWeight: 500 } } },
805   dataLabels: { enabled: false },
806   legend: { position: 'top', fontSize: '12px', markers: { radius: 4 } },
807   grid: { borderColor: '#f1f5f9', strokeDashArray: 4 },
808   tooltip: { y: { formatter: (value) => formatPct(value) } },
809 });
810
811 let prodiChart = null;
812 let kinerjaChart = null;
813 let prodiFullChart = null;
814 let kinerjaFullChart = null;
815 let kepuasanStackChart = null;
816 const compactProdi = sliceData(respondenProdiLabels, respondenProdiData);
817 const compactKinerja = sliceData(chartLabels, chartData);
818
819 setActiveKategori(initialKategori);
820
821 const renderChart = (chart, selector, options) => {
822   if (chart) chart.destroy();
823
824   const target = document.querySelector(selector);
825   if (!target) return null;
826
827   target.innerHTML = '';
828   const nextChart = new ApexCharts(target, options);
829   nextChart.render();
830
831   return nextChart;
832 };
833
834 const renderProdiSummary = () => {
835   setPanelText(
836     'prodiPanelTitle',
837     'prodiPanelSubtitle',
838     'Responden Berdasarkan Program Studi',
839     'Klik bar untuk melihat detail jenis perusahaan pada prodi tersebut'
840   );
841   setBackVisible('backProdiChart', false);

```

```

842
843     if (compactProdi.data.length) {
844         prodiChart = renderChart(prodiChart, '#chart-prodi', prodiOptions(compactProdi.labels, compactProdi.data, false, renderProdiDetail));
845     } else {
846         emptyChart('#chart-prodi', 'Belum ada data responden per Prodi.');
```

```
847     }
848 };
849
```

```
850 const renderProdiDetail = (prodi) => {
```

```
851     const detail = prodiDetailMap.get(prodi);
```

```
852     if (!detail) return;
```

```
853
```

```
854     setPanelText(
```

```
855         'prodiPanelTitle',
```

```
856         'prodiPanelSubtitle',
```

```
857         `Detail ${detail.prodi}`,
```

```
858         `${detail.total} responden | Fakultas: ${detail.fakultas}`
```

```
859     );
```

```
860     setBackVisible('backProdiChart', true);
```

```
861     prodiChart = renderChart(prodiChart, '#chart-prodi', prodiDetailOptions(detail));
```

```
862 };
863
```

```
864 const renderKinerjaSummary = () => {
```

```
865     setPanelText(
```

```
866         'kinerjaPanelTitle',
```

```
867         'kinerjaPanelSubtitle',
```

```
868         'Rata-Rata Penilaian per Kategori',
```

```
869         'Klik bar untuk melihat distribusi jawaban kategori tersebut'
```

```
870     );
```

```
871     setBackVisible('backKinerjaChart', false);
```

```
872
```

```
873     if (compactKinerja.data.length) {
```

```
874         kinerjaChart = renderChart(kinerjaChart, '#chart-kinerja', kinerjaOptions(compactKinerja.labels, compactKinerja.data, false, renderKategoriDetail));
```

```
875     } else {
```

```
876         emptyChart('#chart-kinerja', 'Belum ada data penilaian.');
```

```
877     }
878 };
879
```

```
880 const renderKategoriDetail = (kategori) => {
```

```
881     const detail = kategoriDetailMap.get(kategori);
```

```
882     if (!detail) return;
```

```
883
```

```
884     setActiveKategori(kategori);
```

```
885     setPanelText(
```

```
886         'kinerjaPanelTitle',
```

```
887         'kinerjaPanelSubtitle',
```

```
888         `Detail ${detail.kategori}`,
```

```
889         `Skor rata-rata ${Number(detail.rata_rata || 0).toFixed(2)} dari ${detail.total_respon || 0} respon penilaian`
```

```
890     );
```

```
891     setBackVisible('backKinerjaChart', true);
```

```
892     kinerjaChart = renderChart(kinerjaChart, '#chart-kinerja', kategoriDetailOptions(detail));
```

```
893 };
894
```

```

895 const renderProdiFullSummary = () => {
896   const title = document.getElementById('prodiChartModallabel');
897   if (title) title.textContent = 'Responden Berdasarkan Program Studi';
898   setBackVisible('backProdiFullChart', false);
899
900   if (respondenProdiData.length) {
901     prodiFullChart = renderChart(
902       prodiFullChart,
903       '#chart-prodi-full',
904       prodiOptions(respondenProdiLabels, respondenProdiData, true, renderProdiFullDetail)
905     );
906   } else {
907     emptyChart('#chart-prodi-full', 'Belum ada data responden per Prodi.');
```

```

908   }
909 };
910
911 const renderProdiFullDetail = (prodi) => {
912   const detail = prodiDetailMap.get(prodi);
913   if (!detail) return;
914
915   const title = document.getElementById('prodiChartModallabel');
916   if (title) title.textContent = `Detail ${detail.prodi} - ${detail.total} responden`;
917   setBackVisible('backProdiFullChart', true);
918   prodiFullChart = renderChart(prodiFullChart, '#chart-prodi-full', prodiDetailOptions(detail, true));
919 };
920
921 const renderKinerjaFullSummary = () => {
922   const title = document.getElementById('kinerjaChartModallabel');
923   if (title) title.textContent = 'Rata-Rata Penilaian per Kategori';
924   setBackVisible('backKinerjaFullChart', false);
925
926   if (chartData.length) {
927     kinerjaFullChart = renderChart(
928       kinerjaFullChart,
929       '#chart-kinerja-full',
930       kinerjaOptions(chartLabels, chartData, true, renderKinerjaFullDetail)
931     );
932   } else {
933     emptyChart('#chart-kinerja-full', 'Belum ada data penilaian.');
```

```

934   }
935 };
936
937 const renderKinerjaFullDetail = (kategori) => {
938   const detail = kategoriDetailMap.get(kategori);
939   if (!detail) return;
940
941   setActiveKategori(kategori);
942   const title = document.getElementById('kinerjaChartModallabel');
943   if (title) title.textContent = `Detail ${detail.kategori} - skor ${Number(detail.rata_rata || 0).toFixed(2)}`;
944   setBackVisible('backKinerjaFullChart', true);
945   kinerjaFullChart = renderChart(kinerjaFullChart, '#chart-kinerja-full', kategoriDetailOptions(detail, true));
946 };
947

```

```

948 renderProdiSummary();
949 renderKinerjaSummary();
950
951 document.getElementById('prodiChartModal').addEventListener('shown.bs.modal', () => {
952     if (!prodiFullChart) renderProdiFullSummary();
953 });
954
955 document.getElementById('kinerjaChartModal').addEventListener('shown.bs.modal', () => {
956     if (!kinerjaFullChart) renderKinerjaFullSummary();
957 });
958
959 document.getElementById('backProdiChart')?.addEventListener('click', renderProdiSummary);
960 document.getElementById('backKinerjaChart')?.addEventListener('click', renderKinerjaSummary);
961 document.getElementById('backProdiFullChart')?.addEventListener('click', renderProdiFullSummary);
962 document.getElementById('backKinerjaFullChart')?.addEventListener('click', renderKinerjaFullSummary);
963
964 document.querySelectorAll('[data-kategori-row]').forEach((row) => {
965     row.addEventListener('click', () => {
966         renderKategoriDetail(row.dataset.kategoriRow);
967     });
968 });
969
970 document.querySelectorAll('[data-view-mode]').forEach((button) => {
971     button.addEventListener('click', () => {
972         const mode = button.dataset.viewMode;
973         const panel = button.closest('.satisfaction-panel');
974
975         button.parentElement.querySelectorAll('button').forEach((item) => item.classList.toggle('active', item === button));
976         panel.classList.toggle('chart-mode', mode === 'chart');
977
978         if (mode === 'chart' && !kepuasanStackChart && kategoriDetails.length) {
979             kepuasanStackChart = new ApexCharts(document.querySelector('#chart-kepuasan-stack'), kepuasanStackOptions());
980             kepuasanStackChart.render();
981         }
982     });
983 });
984 </script>
985 @endsection
986

```

5. Controller Survei

File path: app/Http/Controllers/SurveyController.php

Controller untuk pembuatan survei, verifikasi kode akses, penayangan formulir, pengiriman jawaban, pembaruan, dan pembuatan survei massal.

```

1  <?php
2
3  namespace App\Http\Controllers;
4
5  use App\Http\Requests\SurveyBulkRequest;
6  use App\Http\Requests\SurveyStoreRequest;
7  use App\Http\Requests\SurveySubmitJawabanRequest;
8  use App\Http\Requests\SurveyUpdateRequest;
9  use App\Models\Lulusan;
10 use App\Models\PenggunaLulusan;
11 use App\Models\ResponJawaban;
12 use App\Models\soal;
13 use App\Models\Survey;
14 use App\Services\SurveyService;
15 use Illuminate\Http\Request;
16 use Illuminate\Support\Facades\DB;
17 use Illuminate\Support\Str;
18
19 class SurveyController extends Controller
20 {
21     protected $surveyService;
22
23     public function __construct(SurveyService $surveyService)
24     {
25         $this->surveyService = $surveyService;
26     }
27
28     public function index()
29     {
30         // Mengambil semua data survey, diurutkan dari yang terbaru (latest)
31         // with() digunakan untuk memuat relasi (Eager Loading) agar tidak terjadi N+1 problem dan query lebih cepat
32         $surveys = Survey::with(['lulusan', 'penggunaLulusan'])->latest()->get();
33
34         // Mengirim variabel $surveys ke halaman view
35         return view('admin.survey.index', compact('surveys'));
36     }
37
38     public function getPerusahaanData($id)
39     {
40         $data = \App\Models\PenggunaLulusan::find($id);
41         return response()->json($data);
42     }
43     public function add()
44     {
45         $perusahaan = \App\Models\PenggunaLulusan::all();
46

```



```

47 // Ambil semua soal aktif untuk dipilih oleh Admin
48 $daftarSoal = \App\Models\Soal::with('jawaban')
49     ->where('is_active', 1)
50     ->get();
51
52 // Tambahkan data lulusan jika diperlukan di form
53 $lulusan = \App\Models\Lulusan::all();
54
55 return view('admin.survey.add', compact('perusahaan', 'daftarSoal', 'lulusan'));
56 }
57
58 public function store(SurveyStoreRequest $request)
59 {
60     try {
61         $this->surveyService->createSurvey($request->validated());
62         return redirect()->route('survey')->with('success', 'Sesi Survey berhasil dibuat dan data instansi tersinkronisasi.');
```

```

63     } catch (\Exception $e) {
64         return back()->with('error', 'Terjadi kesalahan: ' . $e->getMessage())->withInput();
65     }
66 }
67 public function verifyCode(Request $request)
68 {
69     $survey = Survey::where('access_code', $request->code)
70         ->where('is_completed', false)
71         ->first();
72
73     if (!$survey) {
74         return back()->with('error', 'Kode akses tidak valid atau survey telah selesai.');
```

```

75     }
76
77     return redirect()->route('survey.fill', $survey->access_code);
78 }
79
80 public function fill($code)
81 {
82     $survey = Survey::with(['lulusan', 'penggunaLulusan'])
83         ->where('access_code', $code)
84         ->firstOrFail();
85
86     $fakultasLulusan = $survey->lulusan->fakultas ?? null;
87
88     // Soal yang dipilih admin, difilter berdasarkan peruntukan fakultas lulusan
89     $soal = Soal::whereHas('surveys', function($q) use ($survey) {
90         $q->where('survey_id', $survey->id);
91     })
92     ->where(function($q) use ($fakultasLulusan) {
93         $q->where('peruntukan_fakultas', 'Umum');
94         if ($fakultasLulusan) {
95             $q->orWhere('peruntukan_fakultas', $fakultasLulusan);
96         }
97     })
98     ->with(['jawaban', 'kategori'])
99     ->get();

```

```

100 return view('fill_page', compact('survey', 'soal'));
101 }
102 public function submitJawaban(SurveySubmitJawabanRequest $request, $code)
103 {
104     $survey = Survey::where('access_code', $code)->firstOrFail();
105
106     try {
107         $this->surveyService->submitJawaban($survey, $request->validated());
108
109         return redirect('/')->with('success', 'Terima kasih, kuesioner evaluasi berhasil terkirim dan data Anda telah dicatat!');
110     } catch (\Exception $e) {
111         return back()->with('error', 'Gagal menyimpan jawaban: ' . $e->getMessage())->withInput();
112     }
113 }
114 }
115 public function edit($id)
116 {
117     $survey = Survey::with(['lulusan', 'penggunaLulusan', 'soals.kategori', 'soals.jawaban'])->findOrFail($id);
118
119     $perusahaan = PenggunaLulusan::all();
120     $lulusan = Lulusan::all();
121     $daftarSoal = Soal::where('is_active', 1)->get();
122
123     $responGrouped = $survey->is_completed
124         ? ResponJawaban::with('jawaban')->where('survey_id', $id)->get()->groupBy('soal_id')
125         : collect();
126
127     return view('admin.survey.view', compact('survey', 'perusahaan', 'lulusan', 'daftarSoal', 'responGrouped'));
128 }
129
130 public function bulkCreate()
131 {
132     $tahunList = \App\Models\Lulusan::selectRaw('EXTRACT(YEAR FROM tahun_lulus) as tahun')
133         ->whereNotNull('pengguna_lulusan_id')
134         ->distinct()
135         ->orderByDesc('tahun')
136         ->pluck('tahun');
137
138     $daftarSoal = \App\Models\Soal::with('jawaban')
139         ->where('is_active', 1)
140         ->get();
141
142     return view('admin.survey.bulk', compact('tahunList', 'daftarSoal'));
143 }
144
145 public function bulkStore(SurveyBulkRequest $request)
146 {
147     try {
148         $surveys = $this->surveyService->createBulkSurveys($request->validated());
149         $count = count($surveys);
150         return redirect()->route('survey')->with('success', "Berhasil membuat {$count} survey untuk lulusan tahun {$request->tahun_lulus}.");
151     } catch (\Exception $e) {
152         return back()->with('error', 'Terjadi kesalahan: ' . $e->getMessage())->withInput();
153     }
154 }

```

```

153     }
154 }
155
156 public function getLulusanByTahun(Request $request)
157 {
158     $tahun = $request->tahun;
159     $lulusan = \App\Models\Lulusan::whereRaw('EXTRACT(YEAR FROM tahun_lulus) = ?', [$tahun])
160         ->whereNotNull('pengguna_lulusan_id')
161         ->with('pengguna')
162         ->get(['id', 'nama', 'nim', 'program_studi', 'pengguna_lulusan_id']);
163
164     return response()->json($lulusan);
165 }
166
167 public function update(SurveyUpdateRequest $request, $id)
168 {
169     $survey = Survey::findOrFail($id);
170
171     if ($survey->is_completed) {
172         return back()->with('error', 'Survey sudah diisi dan tidak dapat diubah lagi.');
```

6. Service Pengolahan Survei

File path: app/Services/SurveyService.php

Layanan bisnis untuk membuat survei, menyimpan jawaban secara transaksional, membuat arsip permanen, dan mengelola survei massal.

```

1  <?php
2
3  namespace App\Services;
4
5  use App\Models\lulusan;
6  use App\Models\pengguna_lulusan;
7  use App\Models\ResponJawaban;
8  use App\Models\soal;
9  use App\Models\Survey;
10 use App\Models\SurveyArsip;
11 use Illuminate\Support\Facades\DB;
12 use Illuminate\Support\Str;
13
14 class SurveyService
15 {
16     public function createSurvey(array $data)
17     {
18         return DB::transaction(function () use ($data) {
19             $pengguna = pengguna_lulusan::findOrFail($data['pengguna_lulusan_id']);
20
21             $pengguna->update([
22                 'nama_penyelia' => $data['nama'] ?? $pengguna->nama_penyelia,
23                 'kontak_penyelia' => $data['hp'] ?? $pengguna->kontak_penyelia,
24                 'email_penyelia' => $data['email'] ?? $pengguna->email_penyelia,
25                 'nomor_badan_hukum' => $data['badan_hukum'] ?? $pengguna->nomor_badan_hukum,
26                 'kontak_perusahaan' => $data['telp_perusahaan'] ?? $pengguna->kontak_perusahaan,
27                 'alamat_perusahaan' => $data['alamat_perusahaan'] ?? $pengguna->alamat_perusahaan,
28             ]);
29
30             $lulus = lulusan::findOrFail($data['lulusan_id']);
31
32             $survey = Survey::create([
33                 'judul' => $data['judul'],
34                 'tahun' => $data['tahun'] ?? now()->year,
35                 'deskripsi' => $data['deskripsi'] ?? null,
36                 'lulusan_id' => $lulus->id,
37                 'pengguna_lulusan_id' => $data['pengguna_lulusan_id'],
38                 'access_code' => strtoupper(Str::random(8)),
39                 'is_completed' => false,
40                 'is_active' => true,
41             ]);
42
43             // Hanya simpan soal yang sesuai dengan fakultas lulusan
44             $soalTerpilih = soal::whereIn('id', $data['soal_pilihan'])
45                 ->where(function ($q) use ($lulus) {
46                     $q->where('peruntukan_fakultas', 'Umum');

```

```

47         if ($lulus->fakultas) {
48             $q->orWhere('peruntukan_fakultas', $lulus->fakultas);
49         }
50     })
51     ->pluck('id');
52
53     foreach ($soalTerpilih as $soal_id) {
54         DB::table('survey_soal')->insert([
55             'survey_id' => $survey->id,
56             'soal_id' => $soal_id,
57             'created_at' => now(),
58             'updated_at' => now(),
59         ]);
60     }
61
62     return $survey;
63 });
64 }
65
66 public function submitJawaban(Survey $survey, array $data)
67 {
68     return DB::transaction(function () use ($survey, $data) {
69         if ($survey->pengguna_lulusan_id) {
70             $pengguna = penggunalulusan::find($survey->pengguna_lulusan_id);
71             if ($pengguna) {
72                 $pengguna->update([
73                     'nama_penyelia' => $data['nama_pengisi'],
74                     'jabatan_penyelia' => $data['jabatan_pengisi'] ?? $pengguna->jabatan_penyelia,
75                     'kontak_penyelia' => $data['hp_pengisi'] ?? $pengguna->kontak_penyelia,
76                     'email_penyelia' => $data['email_pengisi'] ?? $pengguna->email_penyelia,
77                     'nama_perusahaan' => $data['nama_perusahaan'] ?? $pengguna->nama_perusahaan,
78                     'nomor_badan_hukum' => $data['nomor_badan_hukum'] ?? $pengguna->nomor_badan_hukum,
79                     'jenis_perusahaan' => $data['jenis_perusahaan'] ?? $pengguna->jenis_perusahaan,
80                     'alamat_perusahaan' => $data['alamat_perusahaan'] ?? $pengguna->alamat_perusahaan,
81                     'kontak_perusahaan' => $data['kontak_perusahaan'] ?? $pengguna->kontak_perusahaan,
82                     'cabang_kota' => $data['cabang_kota'] ?? $pengguna->cabang_kota,
83                     'cabang_negara' => $data['cabang_negara'] ?? $pengguna->cabang_negara,
84                     'jumlah_lulusan' => $data['jumlah_lulusan_bekerja'] ?? $pengguna->jumlah_lulusan,
85                 ]);
86             }
87         }
88
89         $isFirstRecord = true;
90
91         // Pre-load semua soal dan jawaban yang relevan untuk efisiensi query
92         $soalIds = array_keys(array_merge($data['jawaban'] ?? [], $data['mc'] ?? [], $data['mc_custom'] ?? []));
93         $soalCache = soal::whereIn('id', $soalIds)->get()->keyBy('id');
94         $jawabanCache = \App\Models\Jawaban::whereIn('soal_id', $soalIds)->get()->keyBy('id');
95
96         // Rating & Essay
97         foreach ($data['jawaban'] ?? [] as $soal_id => $isi_jawaban) {
98             $soalModel = $soalCache->get($soal_id);
99             if (!$soalModel) continue;

```

```

100
101     $respon = new ResponJawaban();
102     $respon->survey_id      = $survey->id;
103     $respon->soal_id        = $soal_id;
104     $respon->soal_text_snapshot = $soalModel->soal;
105     $respon->responden      = $data['nama_pengisi'];
106
107     if ($isFirstRecord) {
108         $respon->jumlah_lulusan_bekerja = $data['jumlah_lulusan_bekerja'] ?? null;
109         $isFirstRecord = false;
110     }
111
112     if ($soalModel->jenis_soal === 'essay') {
113         $respon->jawaban_text      = $isi_jawaban;
114         $respon->jawaban_id        = null;
115         $respon->jawaban_text_snapshot = null;
116     } else {
117         $jawabanModel              = $jawabanCache->get($isi_jawaban);
118         $respon->jawaban_id        = $isi_jawaban;
119         $respon->jawaban_text_snapshot = $jawabanModel?->jawaban;
120         $respon->jawaban_text      = null;
121     }
122
123     $respon->save();
124 }
125
126 // Multiple Choice: simpan satu baris per jawaban yang dicentang
127 foreach ($data['mc'] ?? [] as $soal_id => $jawaban_ids) {
128     $soalModel = $soalCache->get($soal_id);
129
130     foreach ($jawaban_ids as $jawaban_id) {
131         $jawabanModel = $jawabanCache->get($jawaban_id);
132
133         $respon = new ResponJawaban();
134         $respon->survey_id      = $survey->id;
135         $respon->soal_id        = $soal_id;
136         $respon->soal_text_snapshot = $soalModel?->soal;
137         $respon->responden      = $data['nama_pengisi'];
138         $respon->jawaban_id      = $jawaban_id;
139         $respon->jawaban_text_snapshot = $jawabanModel?->jawaban;
140
141         if ($isFirstRecord) {
142             $respon->jumlah_lulusan_bekerja = $data['jumlah_lulusan_bekerja'] ?? null;
143             $isFirstRecord = false;
144         }
145
146         $respon->save();
147     }
148 }
149
150 // Multiple Choice: simpan teks "Lainnya" jika diisi
151 foreach ($data['mc_custom'] ?? [] as $soal_id => $custom_text) {
152     if (empty(trim($custom_text ?? ''))) continue;

```

```

153
154         $soalModel = $soalCache->get($soal_id);
155
156         $respon = new ResponJawaban();
157         $respon->survey_id      = $survey->id;
158         $respon->soal_id        = $soal_id;
159         $respon->soal_text_snapshot = $soalModel->soal;
160         $respon->responden      = $data['nama_pengisi'];
161         $respon->jawaban_id      = null;
162         $respon->jawaban_text    = trim($custom_text);
163
164         if ($isFirstRecord) {
165             $respon->jumlah_lulusan_bekerja = $data['jumlah_lulusan_bekerja'] ?? null;
166             $isFirstRecord = false;
167         }
168
169         $respon->save();
170     }
171
172     $survey->update(['is_completed' => true]);
173
174     // Tulis arsip permanen – tidak bergantung FK apapun
175     $this->buatArsip($survey->fresh(['lulusan', 'penggunaLulusan']), $data);
176
177     return $survey;
178 });
179 }
180
181 private function buatArsip(Survey $survey, array $data): void
182 {
183     $lulus      = $survey->lulusan;
184     $pengguna   = $survey->penggunaLulusan;
185
186     // Kumpulkan semua soal yang ada di survey ini beserta relasi jawaban & kategori
187     $soals = $survey->soals()->with(['jawaban', 'kategori'])->get()->keyBy('id');
188
189     // Bangun array jawaban terurut berdasarkan kode soal
190     $jawabanArr = [];
191
192     // Rating & Essay (dari $data['jawaban'])
193     foreach ($data['jawaban'] ?? [] as $soal_id => $isi) {
194         $s = $soals->get($soal_id);
195         if (!$s) continue;
196
197         $entry = [
198             'kode'      => $s->kode,
199             'kategori'  => $s->kategori->nama_kategori,
200             'soal'      => $s->soal,
201             'jenis'     => $s->jenis_soal,
202             'nilai'     => null,
203         ];
204
205         if ($s->jenis_soal === 'essay') {

```

```

206         $entry['jawaban'] = $isi;
207     } else {
208         $pil = $s->jawaban->firstWhere('id', $isi);
209         $entry['jawaban'] = $pil->jawaban;
210         $entry['nilai'] = $pil->nilai;
211     }
212
213     $jawabanArr[$s->kode] = $entry;
214 }
215
216 // Multiple Choice
217 foreach ($data['mc'] ?? [] as $soal_id => $jawaban_ids) {
218     $s = $soals->get($soal_id);
219     if (!$s) continue;
220
221     $pilihan = $s->jawaban->whereIn('id', $jawaban_ids)->pluck('jawaban')->toArray();
222     $jawabanArr[$s->kode] = [
223         'kode' => $s->kode,
224         'kategori' => $s->kategori->nama_kategori,
225         'soal' => $s->soal,
226         'jenis' => $s->jenis_soal,
227         'jawaban' => $pilihan,
228         'nilai' => null,
229     ];
230 }
231
232 // Teks "Lainnya" pada multiple choice
233 foreach ($data['mc_custom'] ?? [] as $soal_id => $custom_text) {
234     if (empty(trim($custom_text ?? ''))) continue;
235     if (isset($jawabanArr[$soals->get($soal_id)->kode])) {
236         $jawabanArr[$soals->get($soal_id)->kode]['jawaban'][] = trim($custom_text);
237     }
238 }
239
240 // Urutkan berdasarkan kode soal (B1, B2, C1, ...)
241 ksort($jawabanArr);
242
243 SurveyArsip::create([
244     'survey_id' => $survey->id,
245     'access_code' => $survey->access_code,
246     'judul' => $survey->judul,
247     'submitted_at' => now(),
248     'tahun_instrumen' => $survey->soals->first()->instrumen_id
249     ? \App\Models\Instrumen::find($survey->soals->first()->instrumen_id)->tahun
250     : null,
251
252     'lulusan_nama' => $lulus->nama,
253     'lulusan_nim' => $lulus->nim,
254     'lulusan_program_studi' => $lulus->program_studi,
255     'lulusan_fakultas' => $lulus->fakultas,
256     'lulusan_tahun_lulus' => $lulus->tahun_lulus
257     ? \Carbon\Carbon::parse($lulus->tahun_lulus)->format('Y')
258     : null,

```



```

259
260     'perusahaan_nama'          => $pengguna->nama_perusahaan,
261     'perusahaan_jenis'         => $pengguna->jenis_perusahaan,
262     'perusahaan_alamat'        => $pengguna->alamat_perusahaan,
263     'perusahaan_kontak'        => $pengguna->kontak_perusahaan,
264     'perusahaan_nomor_badan_hukum' => $pengguna->nomor_badan_hukum,
265     'perusahaan_cabang_kota'    => $pengguna->cabang_kota,
266     'perusahaan_cabang_negara'  => $pengguna->cabang_negara,
267
268     'penyelia_nama'            => $data['nama_pengisi'],
269     'penyelia_jabatan'         => $data['jabatan_pengisi'] ?? $pengguna->jabatan_penyelia,
270     'penyelia_email'           => $data['email_pengisi'] ?? $pengguna->email_penyelia,
271     'penyelia_kontak'          => $data['hp_pengisi'] ?? $pengguna->kontak_penyelia,
272     'jumlah_lulusan_bekerja'   => (string) ($data['jumlah_lulusan_bekerja'] ?? null),
273
274     'jawaban_json' => array_values($jawabanArr),
275 ];
276 }
277
278 public function createBulkSurveys(array $data): array
279 {
280     return DB::transaction(function () use ($data) {
281         $tahunLulus = $data['tahun_lulus'];
282
283         $lulusanList = lulusan::whereRaw('EXTRACT(YEAR FROM tahun_lulus) = ?', [$tahunLulus])
284             ->whereNotNull('pengguna_lulusan_id')
285             ->get();
286
287         if ($lulusanList->isEmpty()) {
288             throw new \Exception("Tidak ada lulusan dengan tahun lulus {$tahunLulus} yang memiliki data perusahaan.");
289         }
290
291         $surveys = [];
292
293         // Ambil detail soal yang dipilih admin (termasuk peruntukan_fakultas)
294         $soalTerpilih = soal::whereIn('id', $data['soal_pilihan'])->get(['id', 'peruntukan_fakultas']);
295
296         foreach ($lulusanList as $lulus) {
297             // Filter soal: hanya yang Umum atau sesuai fakultas lulusan ini
298             $soalUntukLulusan = $soalTerpilih->filter(function ($s) use ($lulus) {
299                 return $s->peruntukan_fakultas === 'Umum'
300                     || $s->peruntukan_fakultas === $lulus->fakultas;
301             });
302
303             $survey = Survey::create([
304                 'judul'          => $data['judul'],
305                 'tahun'           => $data['tahun'] ?? now()->year,
306                 'deskripsi'       => $data['deskripsi'] ?? null,
307                 'lulusan_id'      => $lulus->id,
308                 'pengguna_lulusan_id' => $lulus->pengguna_lulusan_id,
309                 'access_code'     => strtoupper(Str::random(8)),
310                 'is_completed'    => false,
311                 'is_active'       => true,

```

```

312     });
313
314     foreach ($soalUntukLulusan as $s) {
315         DB::table('survey_soal')->insert([
316             'survey_id' => $survey->id,
317             'soal_id' => $s->id,
318             'created_at' => now(),
319             'updated_at' => now(),
320         ]);
321     }
322
323     $surveys[] = $survey;
324 }
325
326 return $surveys;
327 });
328 }
329
330 public function updateSurvey(Survey $survey, array $data)
331 {
332     return DB::transaction(function () use ($survey, $data) {
333         $pengguna = penggunalulusan::find($data['pengguna_lulusan_id']);
334         if ($pengguna) {
335             $pengguna->update([
336                 'nama_penyelia' => $data['nama'] ?? $pengguna->nama_penyelia,
337                 'kontak_penyelia' => $data['hp'] ?? $pengguna->kontak_penyelia,
338                 'email_penyelia' => $data['email'] ?? $pengguna->email_penyelia,
339                 'nomor_badan_hukum' => $data['badan_hukum'] ?? $pengguna->nomor_badan_hukum,
340                 'kontak_perusahaan' => $data['telp_perusahaan'] ?? $pengguna->kontak_perusahaan,
341                 'alamat_perusahaan' => $data['alamat_perusahaan'] ?? $pengguna->alamat_perusahaan,
342             ]);
343         }
344
345         $lulus = lulus::findOrFail($data['lulus_id']);
346
347         $survey->update([
348             'judul' => $data['judul'],
349             'tahun' => $data['tahun'] ?? $survey->tahun ?? now()->year,
350             'deskripsi' => $data['deskripsi'] ?? null,
351             'lulusan_id' => $lulus->id,
352             'pengguna_lulusan_id' => $data['pengguna_lulusan_id'],
353         ]);
354
355         // Sync hanya soal yang sesuai dengan fakultas lulus
356         $soalValid = soal::whereIn('id', $data['soal_pilihan'])
357             ->where(function ($q) use ($lulus) {
358                 $q->where('peruntukan_fakultas', 'Umum');
359                 if ($lulus->fakultas) {
360                     $q->orWhere('peruntukan_fakultas', $lulus->fakultas);
361                 }
362             })
363             ->pluck('id')
364             ->toArray();

```

```
365
366         $survey->soals()->sync($soalValid);
367
368         return $survey;
369     });
370 }
371 }
372 }
```

7. Validasi Jawaban Survei

File path: app/Http/Requests/SurveySubmitJawabanRequest.php

Validasi masukan responden sebelum data jawaban, informasi perusahaan, dan identitas pengisi diproses oleh aplikasi.

```
1 <?php
2
3 namespace App\Http\Requests;
4
5 use Illuminate\Foundation\Http\FormRequest;
6
7 class SurveySubmitJawabanRequest extends FormRequest
8 {
9     public function authorize(): bool
10     {
11         return true;
12     }
13
14     public function rules(): array
15     {
16         return [
17             'nama_pengisi' => 'required|string|max:255',
18             'jabatan_pengisi' => 'nullable|string|max:255',
19             'hp_pengisi' => 'nullable|string|max:50',
20             'email_pengisi' => 'nullable|email|max:255',
21             'nama_perusahaan' => 'required|string|max:255',
22             'nomor_badan_hukum' => 'nullable|string|max:255',
23             'jenis_perusahaan' => 'nullable|string|max:255',
24             'alamat_perusahaan' => 'nullable|string',
25             'kontak_perusahaan' => 'nullable|string|max:255',
26             'cabang_kota' => 'nullable|integer|min:0',
27             'cabang_negara' => 'nullable|integer|min:0',
28             'jumlah_lulusan_bekerja' => 'nullable|integer|min:0',
29             'jawaban' => 'nullable|array',
30             'mc' => 'nullable|array',
31             'mc.*' => 'nullable|array',
32             'mc.*.*' => 'nullable|integer',
33             'mc_custom' => 'nullable|array',
34             'mc_custom.*' => 'nullable|string|max:1000',
35         ];
36     }
37 }
38
```

8. Formulir Pengisian Survei

File path: resources/views/fill_page.blade.php

Tampilan publik untuk pengisian profil pengguna lulusan dan instrumen penilaian, termasuk rating, pilihan ganda, dan jawaban esai.

```

1 <!DOCTYPE html>
2 <html lang="id">
3 <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width, initial-scale=1.0">
6     <title>{{ $survey->judul }} - Survey Kepuasan</title>
7
8     <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
9     <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.1/font/bootstrap-icons.css">
10    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap" rel="stylesheet">
11
12    <style>
13        :root {
14            --dinamika-maroon: #8B1A2A; /* Warna khas Universitas Dinamika */
15            --dinamika-maroon-dark: #6C0215;
16            --dinamika-gold: #C9A227;
17            --dinamika-light: #FFF5F7;
18        }
19
20        body {
21            font-family: 'Inter', sans-serif;
22            background-color: #f4f7f6;
23            color: #333;
24        }
25
26        /* Container & Cards */
27        .survey-container {
28            max-width: 850px;
29            margin: 0 auto;
30        }
31        .card-custom {
32            background: #ffffff;
33            border-radius: 16px;
34            border: none;
35            box-shadow: 0 4px 20px rgba(0,0,0,0.03);
36            margin-bottom: 2rem;
37            overflow: hidden;
38        }
39
40        /* Header Survey */
41        .survey-header {
42            background: linear-gradient(135deg, var(--dinamika-maroon) 0%, var(--dinamika-maroon-dark) 100%);
43            color: #fff;
44            padding: 3rem 2rem;
45            text-align: center;
46            position: relative;

```

```

47     overflow: hidden;
48 }
49 /* Aksen kotak-kotak dekoratif ala logo Dinamika */
50 .survey-header::before {
51     content: '';
52     position: absolute;
53     top: -20%;
54     right: -10%;
55     width: 200px;
56     height: 200px;
57     background: rgba(255, 255, 255, 0.05);
58     transform: rotate(45deg);
59     z-index: 0;
60 }
61 .survey-header::after {
62     content: '';
63     position: absolute;
64     bottom: -30%;
65     left: -10%;
66     width: 150px;
67     height: 150px;
68     background: rgba(255, 255, 255, 0.05);
69     transform: rotate(45deg);
70     z-index: 0;
71 }
72 .survey-header-content {
73     position: relative;
74     z-index: 1;
75 }
76 .survey-header h2 {
77     font-weight: 800;
78     margin-bottom: 0.5rem;
79     letter-spacing: -0.5px;
80 }
81 .univ-badge {
82     display: inline-block;
83     background-color: rgba(255,255,255,0.15);
84     padding: 0.4rem 1rem;
85     border-radius: 50px;
86     font-size: 0.85rem;
87     font-weight: 600;
88     letter-spacing: 1px;
89     margin-bottom: 1.5rem;
90     border: 1px solid rgba(255,255,255,0.2);
91     text-transform: uppercase;
92 }
93
94 /* Info Box */
95 .info-box {
96     background-color: #f8fafc;
97     border-radius: 12px;
98     padding: 1.5rem;
99     border: 1px solid #e2e8f0;

```

```

100     }
101     .info-label {
102         font-size: 0.75rem;
103         text-transform: uppercase;
104         letter-spacing: 1px;
105         color: #64748b;
106         font-weight: 600;
107         margin-bottom: 0.25rem;
108     }
109     .info-value {
110         font-weight: 600;
111         color: #0f172a;
112         margin-bottom: 1rem;
113     }
114
115     /* Utilities */
116     .text-dinamika { color: var(--dinamika-maroon) !important; }
117     .bg-dinamika-subtle { background-color: var(--dinamika-light) !important; color: var(--dinamika-maroon) !important; }
118
119     /* Lulusan Card */
120     .lulusan-card {
121         background: linear-gradient(135deg, #fff5f6 0%, #fff 60%);
122         border: 1.5px solid #f5c0c8;
123         border-radius: 14px;
124         padding: 1.75rem 1.5rem;
125         text-align: center;
126         position: relative;
127         overflow: hidden;
128     }
129     .lulusan-card::before {
130         content: '';
131         position: absolute;
132         top: 0; left: 0; right: 0;
133         height: 4px;
134         background: linear-gradient(90deg, var(--dinamika-maroon), var(--dinamika-gold));
135     }
136     .lulusan-avatar {
137         width: 64px;
138         height: 64px;
139         background: linear-gradient(135deg, var(--dinamika-maroon), #c0392b);
140         border-radius: 50%;
141         display: flex;
142         align-items: center;
143         justify-content: center;
144         margin: 0 auto 1rem;
145         box-shadow: 0 4px 14px rgba(154,3,30,0.25);
146     }
147     .lulusan-avatar i {
148         font-size: 1.75rem;
149         color: #fff;
150     }
151     .lulusan-name {
152         font-size: 1.15rem;

```

```

153     font-weight: 700;
154     color: #1e293b;
155     margin-bottom: 0.25rem;
156 }
157 .lulusan-nim {
158     font-size: 0.85rem;
159     font-weight: 600;
160     color: var(--dinamika-maroon);
161     letter-spacing: 0.5px;
162     margin-bottom: 1rem;
163 }
164 .lulusan-meta {
165     display: inline-flex;
166     align-items: center;
167     gap: 0.75rem;
168     background-color: #fff;
169     border: 1px solid #f5c0c8;
170     border-radius: 50px;
171     padding: 0.4rem 1.25rem;
172     font-size: 0.82rem;
173     color: #475569;
174     flex-wrap: wrap;
175     justify-content: center;
176 }
177 .lulusan-meta-item {
178     display: flex;
179     align-items: center;
180     gap: 0.35rem;
181     font-weight: 500;
182 }
183 .lulusan-meta-item i {
184     color: var(--dinamika-maroon);
185     font-size: 0.85rem;
186 }
187 .lulusan-meta-divider {
188     width: 1px;
189     height: 14px;
190     background-color: #f5c0c8;
191 }
192
193 /* Kategori Title */
194 .kategori-title {
195     font-weight: 700;
196     color: #1e293b;
197     background-color: #f1f5f9;
198     padding: 1rem 1.5rem;
199     border-radius: 10px;
200     border-left: 5px solid var(--dinamika-maroon);
201 }
202
203 /* Form Elements */
204 .form-control {
205     border: 1.5px solid #cbd5e1;

```



```

206     border-radius: 10px;
207     padding: 0.75rem 1rem;
208     transition: all 0.2s;
209 }
210 .form-control:focus {
211     border-color: var(--dinamika-maroon);
212     box-shadow: 0 0 4px rgba(154, 3, 30, 0.1);
213 }
214
215 /* Survey Table */
216 .survey-table {
217     border-collapse: collapse;
218     font-size: 0.875rem;
219 }
220 .survey-table thead th {
221     background-color: #fce8eb;
222     color: #7a0117;
223     text-align: center;
224     vertical-align: middle;
225     font-weight: 700;
226     border-color: #f5c0c8;
227     padding: 0.6rem 0.75rem;
228 }
229 .survey-table thead tr:first-child th {
230     background-color: var(--dinamika-maroon);
231     color: #fff;
232     font-size: 0.95rem;
233     text-align: left;
234     letter-spacing: 0.3px;
235     border-color: var(--dinamika-maroon-dark);
236 }
237 .survey-table tbody tr td {
238     vertical-align: middle;
239     border-color: #f5c0c8;
240     padding: 0.6rem 0.75rem;
241 }
242 .survey-table tbody tr:hover td {
243     background-color: #fff5f6;
244 }
245 .survey-table .td-no {
246     text-align: center;
247     font-weight: 600;
248     color: var(--dinamika-maroon);
249     width: 45px;
250 }
251 .survey-table .td-option {
252     text-align: center;
253     width: 65px;
254 }
255 .survey-table .td-option input[type="radio"] {
256     width: 18px;
257     height: 18px;
258     cursor: pointer;

```

```

259     accent-color: var(--dinamika-maroon);
260 }
261 .survey-table .th-option {
262     min-width: 60px;
263     line-height: 1.3;
264 }
265 .survey-table .th-option .score-badge {
266     display: block;
267     font-size: 0.7rem;
268     font-weight: 400;
269     opacity: 0.75;
270 }
271
272 /* Multiple Choice */
273 .mc-option .form-check-input {
274     width: 18px;
275     height: 18px;
276     cursor: pointer;
277     accent-color: var(--dinamika-maroon);
278 }
279 .mc-option .form-check-label {
280     font-size: 0.9rem;
281     cursor: pointer;
282     padding-top: 1px;
283 }
284 .mc-other-input {
285     border: 1.5px solid #cbd5e1;
286     border-radius: 8px;
287     padding: 0.45rem 0.85rem;
288     font-size: 0.875rem;
289     transition: all 0.2s;
290 }
291 .mc-other-input:focus {
292     border-color: var(--dinamika-maroon);
293     box-shadow: 0 0 3px rgba(154, 3, 30, 0.1);
294     outline: none;
295 }
296 .mc-other-input:disabled {
297     background-color: #f8fafc;
298     cursor: not-allowed;
299 }
300
301 /* Button */
302 .btn-submit {
303     background-color: var(--dinamika-maroon);
304     color: white;
305     border: none;
306     border-radius: 10px;
307     padding: 1rem 2rem;
308     font-weight: 600;
309     font-size: 1.1rem;
310     width: 100%;
311     transition: all 0.3s;

```

```

312     text-transform: uppercase;
313     letter-spacing: 1px;
314 }
315 .btn-submit:hover {
316     background-color: var(--dinamika-maroon-dark);
317     transform: translateY(-2px);
318     box-shadow: 0 8px 20px rgba(154, 3, 30, 0.2);
319     color: white;
320 }
321
322 .border-bottom-dashed { border-bottom: 2px dashed #e2e8f0; }
323 </style>
324 </head>
325 <body>
326
327 <div class="container py-5 survey-container">
328
329     @if(session('error'))
330     <div class="alert alert-danger shadow-sm rounded-3 border-0 mb-4">
331         <i class="bi bi-exclamation-triangle-fill me-2"></i> {{ session('error') }}
332     </div>
333     @endif
334
335     <div class="card-custom">
336         <div class="survey-header">
337             <div class="survey-header-content">
338                 <div class="univ-badge"><i class="bi bi-mortarboard-fill me-2 text-warning"></i> Universitas Dinamika</div>
339                 <h2>{{ $survey->judul }}</h2>
340                 <p class="text-white-50 mb-0">Kode Akses Sesi: <span class="fw-bold text-white tracking-widest">{{ $survey->access_code }}</span></p>
341             </div>
342         </div>
343
344         <div class="p-4 p-md-5">
345             @if($survey->deskripsi)
346                 <div class="alert alert-light border mb-4 text-secondary" style="line-height: 1.6;">
347                     <i class="bi bi-info-circle-fill text-dinamika me-2"></i> {{ $survey->deskripsi }}
348                 </div>
349             @endif
350
351             <h5 class="fw-bold mb-4 text-dark">A. Informasi tentang Lulusan Terkait</h5>
352
353             <div class="lulusan-card">
354                 <div class="lulusan-avatar">
355                     <i class="bi bi-person-badge-fill"></i>
356                 </div>
357                 <div class="lulusan-name">{{ $survey->lulusan->nama ?? '-' }}</div>
358                 <div class="lulusan-nim">{{ $survey->lulusan->nim ?? '-' }}</div>
359                 <div class="lulusan-meta">
360                     <div class="lulusan-meta-item">
361                         <i class="bi bi-building-fill-check"></i>
362                         <span>{{ $survey->lulusan->program_studi ?? '-' }}</span>
363                     </div>
364                 <div class="lulusan-meta-divider"></div>

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```

365         <div class="lulusan-meta-item">
366             <i class="bi bi-mortarboard-fill"></i>
367             <span>Lulus {{ $survey->lulusan->tahun_lulus ? \Carbon\Carbon::parse($survey->lulusan->tahun_lulus)->format('Y') : '-' }}</span>
368         </div>
369     </div>
370 </div>
371 </div>
372 </div>
373
374 <form action="{{ route('survey.submit', $survey->access_code) }}" method="POST">
375     @csrf
376
377     <div class="card-custom p-4 p-md-5">
378         <div class="d-flex justify-content-between align-items-center mb-2">
379             <div>
380                 <label class="form-label fs-5 fw-bold text-dark mb-0">Identitas Perusahaan <span class="text-danger">*</span></label>
381                 <p class="text-muted small mb-0 mt-1">Data institusi/perusahaan Anda sudah terisi dari sistem.</p>
382             </div>
383         </div>
384
385         @php
386             $autoJumlah = $survey->penggunalulusan->lulusans->count();
387             $storedJumlah = $survey->penggunalulusan->jumlah_lulusan;
388             $defaultJumlah = $storedJumlah ?? $autoJumlah;
389         @endphp
390
391         <div class="p-3 bg-light rounded border mb-3">
392             <div class="d-flex align-items-center">
393                 <i class="bi bi-building fs-3 text-secondary me-3"></i>
394                 <div class="flex-grow-1">
395                     <div class="fw-bold">{{ $survey->penggunalulusan->nama_perusahaan ?? 'Nama Perusahaan Belum Tersedia' }}</div>
396                     <div class="small text-muted">{{ $survey->penggunalulusan->alamat_perusahaan ?? 'Alamat belum tersedia' }}</div>
397                 </div>
398                 <div class="text-end ms-3">
399                     <div class="fw-bold text-primary fs-4">{{ $defaultJumlah }}</div>
400                     <div class="small text-muted">lulusan tercatat</div>
401                 </div>
402             </div>
403         </div>
404
405         <div class="mb-3 p-3 border rounded bg-white">
406             <label class="form-label small text-secondary fw-bold mb-1">
407                 <i class="bi bi-people-fill me-1 text-primary"></i>
408                 Jumlah Lulusan yang Pernah Bekerja di Instansi Ini
409             </label>
410             <div class="d-flex align-items-center gap-2">
411                 <input type="number" name="jumlah_lulusan_bekerja" class="form-control"
412                     style="max-width: 160px"
413                     min="0" value="{{ $defaultJumlah }}"
414                     placeholder="Jumlah orang...">
415                 @if($autoJumlah > 0)
416                 <span class="text-muted small">
417                     (sistem mencatat <strong>{{ $autoJumlah }}</strong> lulusan)

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418         </span>
419     @endif
420 </div>
421 <div class="form-text">Ubah jika jumlah sebenarnya berbeda dari data sistem.</div>
422 </div>
423
424 <button class="btn btn-outline-secondary btn-sm" type="button" data-bs-toggle="collapse" data-bs-target="#collapsePerusahaan" aria-expanded="false" aria-
controls="collapsePerusahaan">
425     <i class="bi bi-pencil-square me-1"></i> Ubah Data Perusahaan Jika Tidak Sesuai
426 </button>
427
428 <div class="collapse mt-3" id="collapsePerusahaan">
429     <div class="row g-3 border-top pt-3 mt-1">
430         <div class="col-md-12">
431             <label class="form-label small text-secondary fw-bold mb-1">Nama Perusahaan <span class="text-danger">*</span></label>
432 >nama_perusahaan ?? ' ' </div>
433         </div>
434         <div class="col-md-6">
435             <label class="form-label small text-secondary fw-bold mb-1">Nomor Badan Hukum</label>
436 >nomor_badan_hukum ?? ' ' </div>
437         </div>
438         <div class="col-md-6">
439             <label class="form-label small text-secondary fw-bold mb-1">Jenis Perusahaan</label>
440 @php
441     $jenisOptions = [
442         'Teknologi Informasi / Software / Digital',
443         'Keuangan / Perbankan / Bisnis',
444         'Manufaktur / Industri',
445         'Perdagangan / Retail / E-Commerce',
446         'Media / Kreatif / Desain',
447         'Jasa (konsultan, pendidikan, dll)',
448         'Pemerintahan',
449         'BUMN / BUMD',
450     ];
451     $currentJenis = $survey->penggunalulusan->jenis_perusahaan ?? '';
452     $isLainnya = $currentJenis !== '' && !in_array($currentJenis, $jenisOptions);
453 @endphp
454 <select name="jenis_perusahaan" id="fill_jenis_select" class="form-control"
455     onchange="document.getElementById('fill_jenis_lainnya_wrap').classList.toggle('d-none',this.value!=='_lainnya_')">
456     <option value="" disabled {{ ! $currentJenis ? 'selected' : '' }}>Pilih Jenis Perusahaan</option>
457 >>Teknologi Informasi / Software / Digital {{ $currentJenis === 'Teknologi Informasi / Software / Digital' ? 'selected' : '' }}</option>
458 / Bisnis</option>
459 >>Manufaktur / Industri {{ $currentJenis === 'Manufaktur / Industri' ? 'selected' : '' }}</option>
460 >>Perdagangan / Retail / E-Commerce {{ $currentJenis === 'Perdagangan / Retail / E-Commerce' ? 'selected' : '' }}</option>
461 >>Media / Kreatif / Desain {{ $currentJenis === 'Media / Kreatif / Desain' ? 'selected' : '' }}</option>
462 >>Jasa (konsultan, pendidikan, dll) {{ $currentJenis === 'Jasa (konsultan, pendidikan, dll)' ? 'selected' : '' }}</option>
463 >>Pemerintahan {{ $currentJenis === 'Pemerintahan' ? 'selected' : '' }}</option>

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464         <option value="BUMN / BUMD" {{ $currentJenis === 'BUMN / BUMD' ? 'selected' : '' }}>BUMN / BUMD</option>
465         <option value="_lainnya_" {{ $isLainnya ? 'selected' : '' }}>Lainnya...</option>
466     </select>
467     <div id="fill_jenis_lainnya_wrap" class="mt-1 {{ $isLainnya ? '' : 'd-none' }}">
468         <input type="text" id="fill_jenis_lainnya_text" class="form-control form-control-sm"
469             placeholder="Sebutkan jenis perusahaan..."
470             value="{{ $isLainnya ? $currentJenis : '' }}"
471             oninput="document.getElementById('fill_jenis_select').value='_lainnya_'"
472     </div>
473 </div>
474 <script>
475 document.querySelector('form').addEventListener('submit', function(e) {
476     var sel = document.getElementById('fill_jenis_select');
477     var text = document.getElementById('fill_jenis_lainnya_text');
478     if (sel && sel.value === '_lainnya_') {
479         if (!text.value.trim()) {
480             e.preventDefault();
481             text.focus();
482             text.classList.add('is-invalid');
483             return;
484         }
485         sel.value = text.value.trim();
486     }
487 });
488 </script>
489 <div class="col-md-12">
490     <label class="form-label small text-secondary fw-bold mb-1">Alamat Perusahaan</label>
491     <textarea name="alamat_perusahaan" class="form-control" rows="2" placeholder="Alamat lengkap...">{{ $survey->penggunalulusan->alamat_perusahaan
492     ?? '' }}</textarea>
493 </div>
494 <div class="col-md-4">
495     <label class="form-label small text-secondary fw-bold mb-1">Kontak Perusahaan</label>
496     <input type="text" name="kontak_perusahaan" class="form-control" placeholder="Telp/WA Perusahaan..." value="{{ $survey->penggunalulusan-
497     >kontak_perusahaan ?? '' }}">
498 </div>
499 <div class="col-md-4">
500     <label class="form-label small text-secondary fw-bold mb-1">Jumlah Cabang Nasional</label>
501     <input type="number" name="cabang_kota" class="form-control" min="0" placeholder="Jumlah cabang..." value="{{ $survey->penggunalulusan-
502     >cabang_kota ?? 0 }}">
503 </div>
504 <div class="col-md-4">
505     <label class="form-label small text-secondary fw-bold mb-1">Jumlah Cabang Luar Negeri</label>
506     <input type="number" name="cabang_negara" class="form-control" min="0" placeholder="Jumlah negara..." value="{{ $survey->penggunalulusan-
507     >cabang_negara ?? 0 }}">
508 </div>
509 </div>
510 </div>
511 <div class="card-custom p-4 p-md-5">
512     <div class="mb-2">
513         <label class="form-label fs-5 fw-bold text-dark">Konfirmasi Identitas Responden <span class="text-danger">*</span></label>
514         <p class="text-muted small mb-3">Mohon konfirmasi atau lengkapi data Anda sebagai perwakilan instansi yang mengisi kuesioner ini.</p>

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513         </div>
514
515         <div class="row g-3">
516             <div class="col-md-6">
517                 <label class="form-label small text-secondary fw-bold mb-1">Nama Lengkap Penyelia <span class="text-danger">*/</span></label>
518                 <input type="text" name="nama_pengisi" class="form-control" required placeholder="Nama Anda..." value="{{ $survey->penggunalulusan->nama_penyelid ?? '' }}">
519             </div>
520             <div class="col-md-6">
521                 <label class="form-label small text-secondary fw-bold mb-1">Jabatan / Posisi</label>
522                 <input type="text" name="jabatan_pengisi" class="form-control" placeholder="Contoh: HRD Manager, Direktur..." value="{{ $survey->penggunalulusan->jabatan_penyelid ?? '' }}">
523             </div>
524             <div class="col-md-6">
525                 <label class="form-label small text-secondary fw-bold mb-1">Nomor HP</label>
526                 <input type="text" name="hp_pengisi" class="form-control" placeholder="08..." value="{{ $survey->penggunalulusan->kontak_penyelid ?? '' }}">
527             </div>
528             <div class="col-md-6">
529                 <label class="form-label small text-secondary fw-bold mb-1">Alamat Email</label>
530                 <input type="email" name="email_pengisi" class="form-control" placeholder="email@instansi.com" value="{{ $survey->penggunalulusan->email_penyelid ?? '' }}">
531             </div>
532         </div>
533     </div>
534
535     @php
536         $groupedSoal = $soal->groupBy(fn($s) => $s->kategori->nama_kategori ?? 'Lainnya');
537     @endphp
538
539     @foreach($groupedSoal as $namaKategori => $soalGroup)
540         @php
541             $ratingSoal = $soalGroup->filter(fn($s) => $s->jenis_soal === 'rating')->values();
542             $multiChoiceSoal = $soalGroup->filter(fn($s) => $s->jenis_soal === 'multiple_choice')->values();
543             $essaySoal = $soalGroup->filter(fn($s) => $s->jenis_soal === 'essay')->values();
544             $firstRating = $ratingSoal->first();
545         @endphp
546
547         <div class="card-custom p-4 p-md-5">
548
549             {{-- RATING: tabel radio button, pilih salah satu --}}
550             @if($ratingSoal->count() > 0)
551                 <div class="table-responsive">
552                     <table class="table table-bordered survey-table mb-0">
553                         <thead>
554                             <tr>
555                                 <th colspan="{{ 2 + ($firstRating ? $firstRating->jawaban->count() : 0) }}">
556                                     {{ $namaKategori }}
557                                 </th>
558                             </tr>
559                             <tr>
560                                 <th class="td-no">No</th>
561                                 <th>Pertanyaan</th>
562                                 @if($firstRating)

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```

563         @foreach($firstRating->jawaban as $j)
564             <th class="th-option">
565                 {{ $j->jawaban }}
566                 @if($j->nilai)
567                     <span class="score-badge">{{ $j->nilai }}</span>
568                 @endif
569             </th>
570         @endforeach
571     @endif
572 </tr>
573 </thead>
574 <tbody>
575     @foreach($ratingSoal as $s)
576         <tr>
577             <td class="td-no">{{ $loop->iteration }}</td>
578             <td>
579                 {{ $s->soal }}
580                 @if($s->is_required) <span class="text-danger">*</span> @endif
581             </td>
582             @foreach($s->jawaban as $j)
583                 <td class="td-option">
584                     <input type="radio"
585                         name="jawaban[{{ $s->id }}]"
586                         value="{{ $j->id }}"
587                         {{ $s->is_required ? 'required' : '' }}>
588                 </td>
589             @endforeach
590         </tr>
591     @endforeach
592 </tbody>
593 </table>
594 </div>
595 @endif
596
597 {{-- MULTIPLE CHOICE: checkbox beberapa pilihan + kolom "Lainnya" teks bebas --}}
598 @if($multiChoiceSoal->count() > 0)
599     <div class="{{ $ratingSoal->count() > 0 ? 'mt-4 pt-4 border-top' : '' }}">
600         @if($ratingSoal->count() == 0)
601             <div class="kategori-title mb-4">
602                 <h5 class="mb-0">{{ $namaKategori }}</h5>
603             </div>
604         @endif
605
606         @foreach($multiChoiceSoal as $s)
607             <div class="mb-4 {{ !$loop->last ? 'pb-4 border-bottom-dashed' : '' }}">
608                 <p class="fw-bold text-dark mb-3">
609                     <span class="text-dinamika me-1">{{ $loop->iteration }}</span>
610                     {{ $s->soal }}
611                     @if($s->is_required) <span class="text-danger">*</span> @endif
612                 </p>
613                 <div class="ps-2">
614                     @foreach($s->jawaban as $j)
615                         <div class="form-check mc-option mb-2">

```



```

616         <input class="form-check-input" type="checkbox"
617             name="mc[{{ $s->id }}][]"
618             value="{{ $j->id }}"
619             id="mc_{{ $s->id }}_{{ $j->id }}">
620         <label class="form-check-label" for="mc_{{ $s->id }}_{{ $j->id }}">
621             {{ $j->jawaban }}
622             @if($j->nilai)
623                 <span class="text-muted small">{{ $j->nilai }}</span>
624             @endif
625         </label>
626     </div>
627 @endforeach
628
629 <div class="form-check mc-option mb-2">
630     <input class="form-check-input" type="checkbox"
631         id="mc_other_check_{{ $s->id }}"
632         onchange="
633             var txt = document.getElementById('mc_other_text_{{ $s->id }}');
634             txt.disabled = !this.checked;
635             if (this.checked) txt.focus();
636             else txt.value = '';
637         ">
638     <label class="form-check-label" for="mc_other_check_{{ $s->id }}">
639         Lainnya (tuliskan):
640     </label>
641 </div>
642 <input type="text"
643     name="mc_custom[{{ $s->id }}]"
644     id="mc_other_text_{{ $s->id }}"
645     class="mc-other-input ms-4 d-block"
646     style="max-width: 420px"
647     placeholder="Tuliskan jawaban Anda..."
648     disabled>
649 </div>
650 </div>
651 @endforeach
652 </div>
653 @endif
654
655 {{-- ESSAY: textarea teks bebas --}}
656 @if($essaySoal->count() > 0)
657     @php $hasSectionAbove = $ratingSoal->count() > 0 || $multiChoiceSoal->count() > 0; @endphp
658     <div class="{{ $hasSectionAbove ? 'mt-4 pt-4 border-top' : '' }}">
659         @if(!$hasSectionAbove)
660             <div class="kategori-title mb-4">
661                 <h5 class="mb-0">{{ $namaKategori }}</h5>
662             </div>
663         @endif
664         @foreach($essaySoal as $s)
665             <div class="mb-4 {{ !$loop->last ? 'pb-4 border-bottom-dashed' : '' }}">
666                 <label class="form-label fs-6 mb-2 text-dark fw-bold">
667                     <span class="text-dinamika me-1">{{ $loop->iteration }}</span> {{ $s->soal }}
668                     @if($s->is_required) <span class="text-danger">*</span> @endif

```

```

669         </label>
670         <textarea name="jawaban[{{ $s->id }}" class="form-control" rows="3"
671             placeholder="Tuliskan umpan balik Anda di sini..."
672             {{ $s->is_required ? 'required' : '' }}></textarea>
673     </div>
674     @endforeach
675 </div>
676 @endif
677
678 </div>
679 @endforeach
680
681 <div class="mt-4 mb-5">
682     <button type="submit" class="btn-submit">
683         Kirim Kuesioner Evaluasi <i class="bi bi-send-check ms-2"></i>
684     </button>
685     <p class="text-center text-muted small mt-3">
686         <i class="bi bi-shield-lock-fill text-success"></i> Data umpan balik ini disimpan dengan aman dan rahasia.
687     </p>
688 </div>
689
690 </form>
691 </div>
692
693 <!-- Bootstrap JS Bundle with Popper -->
694 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
695 </body>
696 </html>
697

```

9. Service Pengelolaan Pertanyaan

File path: app/Services/PertanyaanService.php

Logika pengelolaan kategori, jenis pertanyaan, pilihan jawaban, nilai skala, serta status aktif instrumen survei.

```

1  <?php
2
3  namespace App\Services;
4
5  use App\Models\jawaban;
6  use App\Models\soal;
7  use Illuminate\Support\Facades\DB;
8
9  class PertanyaanService
10 {
11     public function storePertanyaan(array $data)
12     {
13         return DB::transaction(function () use ($data) {
14             // 1. Konversi Jenis Soal (Sesuai logic Anda sebelumnya)
15             $jenis = $this->mapJenisSoal($data['type']);
16
17             // 2. Simpan Soal
18             $soal = Soal::create([
19                 'soal'           => $data['question'],
20                 'kategori_id'    => $data['kategori_id'], // Simpan data kategori ke database
21                 'peruntukan_fakultas' => $data['peruntukan_fakultas'],
22                 'kode'           => $data['kode'] ?? null,
23                 'jenis_soal'      => $jenis,
24                 'is_required'     => isset($data['required']),
25                 'is_active'       => true
26             ]);
27
28             // 3. Simpan Jawaban jika Multiple Choice / Radio
29             if ($jenis === 'multiple_choice' && isset($data['jawaban'])) {
30                 $this->saveJawaban($soal->id, $data['jawaban'], $data['nilai'] ?? []);
31             }
32
33             return $soal;
34         });
35     }
36
37     /**
38      * Memperbarui Soal dan Jawaban
39      */
40     public function updatePertanyaan($id, array $data)
41     {
42         return DB::transaction(function () use ($id, $data) {
43             $soal = Soal::findOrFail($id);
44             $jenis = $this->mapJenisSoal($data['type']);
45
46             $soal->update([

```

```

47         'soal'                => $data['question'],
48         'kategori_id'         => $data['kategori_id'], // Simpan kategori yang diupdate
49         'peruntukan_fakultas' => $data['peruntukan_fakultas'],
50         'kode'                => $data['kode'],
51         'jenis_soal'          => $jenis,
52         'is_required'         => isset($data['required']),
53     ]));
54
55     // Hapus jawaban lama dan simpan yang baru
56     $soal->jawaban()->delete();
57
58     if ($jenis === 'multiple_choice' && isset($data['jawaban'])) {
59         $this->saveJawaban($soal->id, $data['jawaban'], $data['nilai'] ?? []);
60     }
61
62     return $soal;
63 });
64 }
65
66 /**
67  * Toggle Status Aktif
68  */
69 public function toggleStatus($id)
70 {
71     $soal = soal::findOrFail($id);
72     $soal->is_active = !$soal->is_active;
73     $soal->save();
74
75     return $soal;
76 }
77
78 // Helper: Pemetaan Jenis
79 private function mapJenisSoal($type)
80 {
81     return match ($type) {
82         'radio' => 'multiple_choice',
83         'text'  => 'essay',
84         default => 'rating',
85     };
86 }
87
88 // Helper: Simpan Jawaban Iterasi
89 private function saveJawaban($soalId, $jawabanList, $nilaiList)
90 {
91     foreach ($jawabanList as $index => $teks) {
92         jawaban::create([
93             'soal_id' => $soalId,
94             'jawaban' => $teks,
95             'nilai'   => $nilaiList[$index] ?? 0,
96             'urutan'  => $index + 1
97         ]);
98     }
99 }

```

100	}
101	

10. Seeder Instrumen Survei Pengguna Lulusan

File path: database/seeder/DraftInstrumenUniversitas2026Seeder.php

Pembentukan instrumen, kategori penilaian, butir pertanyaan, dan bobot jawaban skala empat sebagai dasar pengolahan hasil survei.

```

1  <?php
2
3  namespace Database\Seeders;
4
5  use Carbon\Carbon;
6  use Illuminate\Database\Seeder;
7  use Illuminate\Support\Facades\DB;
8
9  class DraftInstrumenUniversitas2026Seeder extends Seeder
10 {
11     public function run(): void
12     {
13         $tahun = 2026;
14
15         // Tolak jika instrumen tahun ini sudah ada – agar data yang sudah diisi tidak terhapus
16         if (DB::table('instrumen')->where('tahun', $tahun)->exists()) {
17             $this->command->warn("Instrumen tahun {$tahun} sudah ada. Seeder dilewati untuk mencegah penghapusan data.");
18
19             return;
20         }
21
22         // Hanya hapus soal/jawaban yang belum terhubung ke survey manapun (belum ada respon)
23         $soalAman = DB::table('soal')
24             ->whereNull('instrumen_id')
25             ->whereNotIn('id', DB::table('survey_soal')->pluck('soal_id'))
26             ->pluck('id');
27
28         if ($soalAman->isNotEmpty()) {
29             DB::table('jawaban')->whereIn('soal_id', $soalAman)->delete();
30             DB::table('soal')->whereIn('id', $soalAman)->delete();
31         }
32
33         // Hapus kategori yang tidak lagi punya soal
34         DB::table('kategori')
35             ->whereNotIn('id', DB::table('soal')->pluck('kategori_id')->filter())
36             ->delete();
37
38         $likert = [
39             ['jawaban' => 'Sangat Baik', 'nilai' => 4, 'urutan' => 1],
40             ['jawaban' => 'Baik', 'nilai' => 3, 'urutan' => 2],
41             ['jawaban' => 'Cukup', 'nilai' => 2, 'urutan' => 3],
42             ['jawaban' => 'Kurang', 'nilai' => 1, 'urutan' => 4],
43         ];
44
45         $instrument = [
46

```

```

47 // — Halaman 3 —————
48
49 // — Halaman 3-4 —————
50 'B. Etika' => [
51   'deskripsi' => 'Penilaian kejujuran dan profesionalisme lulusan di tempat kerja.',
52   'soal' => [
53     'B1' => ['teks' => 'Lulusan bertindak jujur dalam bekerja',           'jenis' => 'rating', 'pilihan' => $likert],
54     'B2' => ['teks' => 'Lulusan bersikap profesional dalam berinteraksi di tempat kerja', 'jenis' => 'rating', 'pilihan' => $likert],
55   ],
56 ],
57
58 'C. Keahlian Berdasarkan Bidang Ilmu' => [
59   'deskripsi' => 'Kompetensi dan kemampuan lulusan sesuai bidang studi yang ditempuh.',
60   'soal' => [
61     'C1' => ['teks' => 'Kesesuaian kompetensi lulusan dengan kebutuhan pekerjaan',           'jenis' => 'rating', 'pilihan' =>
62     $likert],
63     'C2' => ['teks' => 'Kemampuan lulusan dalam menyelesaikan pekerjaan sesuai bidangnya',           'jenis' => 'rating', 'pilihan' =>
64     $likert],
65     'C3' => ['teks' => 'Kemampuan lulusan dalam menerapkan pengetahuan dan keterampilannya dalam pekerjaan',           'jenis' => 'rating', 'pilihan' =>
66     $likert],
67   ],
68 ],
69
70 'D. Kemampuan Berbahasa Asing' => [
71   'deskripsi' => 'Kemampuan lulusan dalam berkomunikasi menggunakan bahasa asing.',
72   'soal' => [
73     'D1' => ['teks' => 'Kemampuan lulusan dalam memahami komunikasi dalam bahasa asing',           'jenis' => 'rating', 'pilihan' => $likert],
74     'D2' => ['teks' => 'Kemampuan lulusan dalam berkomunikasi menggunakan bahasa asing',           'jenis' => 'rating', 'pilihan' => $likert],
75   ],
76 ],
77
78 'E. Penggunaan Teknologi Informasi' => [
79   'deskripsi' => 'Kemampuan lulusan dalam memanfaatkan dan beradaptasi dengan teknologi.',
80   'soal' => [
81     'E1' => ['teks' => 'Kemampuan lulusan dalam menggunakan teknologi atau aplikasi yang mendukung pekerjaan', 'jenis' => 'rating', 'pilihan' => $likert],
82     'E2' => ['teks' => 'Kemampuan lulusan dalam mempelajari dan beradaptasi dengan teknologi baru',           'jenis' => 'rating', 'pilihan' => $likert],
83   ],
84 ],
85
86 // — Halaman 4 —————
87 'F. Kemampuan Berkomunikasi' => [
88   'deskripsi' => 'Kemampuan lulusan menyampaikan informasi dan berkomunikasi di lingkungan kerja.',
89   'soal' => [
90     'F1' => ['teks' => 'Kemampuan lulusan dalam menyampaikan ide atau informasi dengan jelas',           'jenis' => 'rating', 'pilihan' => $likert],
91     'F2' => ['teks' => 'Kemampuan lulusan dalam berkomunikasi secara efektif di lingkungan kerja',           'jenis' => 'rating', 'pilihan' => $likert],
92   ],
93 ],
94
95 'G. Kerjasama Tim' => [
96   'deskripsi' => 'Kemampuan lulusan bekerja sama dan berkontribusi dalam tim.',
97   'soal' => [
98     'G1' => ['teks' => 'Kemampuan lulusan dalam bekerja sama dalam tim',           'jenis' => 'rating', 'pilihan' => $likert],
99     'G2' => ['teks' => 'Kemampuan lulusan dalam berkontribusi untuk mencapai tujuan tim',           'jenis' => 'rating', 'pilihan' => $likert],

```

```

97     ],
98 ],
99
100 'H. Pengembangan Diri' => [
101   'deskripsi' => 'Kemauan dan kemampuan lulusan untuk terus berkembang dan beradaptasi.',
102   'soal' => [
103     'H1' => ['teks' => 'Kemauan lulusan untuk terus belajar dan meningkatkan kompetensi', 'jenis' => 'rating', 'pilihan' => $likert],
104     'H2' => ['teks' => 'Kemampuan lulusan dalam beradaptasi terhadap perubahan di lingkungan kerja', 'jenis' => 'rating', 'pilihan' => $likert],
105   ],
106 ],
107
108 'I. Kepemimpinan' => [
109   'deskripsi' => 'Kemampuan lulusan mengambil inisiatif dan mengoordinasikan rekan kerja.',
110   'soal' => [
111     'I1' => ['teks' => 'Kemampuan lulusan dalam mengambil inisiatif dalam pekerjaan', 'jenis' => 'rating', 'pilihan' => $likert],
112     'I2' => ['teks' => 'Kemampuan lulusan dalam mengarahkan dan mengoordinasikan rekan kerja', 'jenis' => 'rating', 'pilihan' => $likert],
113   ],
114 ],
115
116 'J. Etos Kerja' => [
117   'deskripsi' => 'Kedisiplinan, tanggung jawab, dan inisiatif lulusan dalam menjalankan tugas.',
118   'soal' => [
119     'J1' => ['teks' => 'Disiplin kerja lulusan dalam menjalankan tugas', 'jenis' => 'rating', 'pilihan' => $likert],
120     'J2' => ['teks' => 'Tanggung jawab lulusan terhadap pekerjaan yang diberikan', 'jenis' => 'rating', 'pilihan' => $likert],
121     'J3' => ['teks' => 'Inisiatif lulusan dalam menyelesaikan pekerjaan', 'jenis' => 'rating', 'pilihan' => $likert],
122   ],
123 ],
124
125 // — Halaman 5 —————
126 'K. Evaluasi Umum' => [
127   'deskripsi' => 'Penilaian umum terhadap keseluruhan kualitas lulusan.',
128   'soal' => [
129     'K1' => [
130       'teks' => 'Secara umum, bagaimana penilaian Anda terhadap kualitas lulusan kami?',
131       'jenis' => 'multiple_choice',
132       'pilihan' => $likert,
133     ],
134   ],
135 ],
136
137 'L. Masukan dan Saran' => [
138   'deskripsi' => 'Masukan kualitatif untuk peningkatan kualitas lulusan.',
139   'soal' => [
140     'L1' => [
141       'teks' => 'Apa yang perlu ditingkatkan dari lulusan kami agar lebih sesuai dengan kebutuhan industri?',
142       'jenis' => 'essay',
143       'pilihan' => [],
144     ],
145   ],
146 ],
147
148 'M. Kerjasama Lanjutan' => [
149   'deskripsi' => 'Potensi bentuk kerja sama antara perusahaan dan kampus.',

```



```

150     'soal' => [
151         'M1' => [
152             'teks' => 'Bentuk kerja sama apa yang berpotensi dapat dilakukan antara perusahaan Anda dengan kampus kami?',
153             'jenis' => 'multiple_choice',
154             'pilihan' => [
155                 ['jawaban' => 'Rekrutmen Lulusan', 'nilai' => 1, 'urutan' => 1],
156                 ['jawaban' => 'Internship / Magang Mahasiswa', 'nilai' => 2, 'urutan' => 2],
157                 ['jawaban' => 'Kuliah tamu / narasumber praktisi', 'nilai' => 3, 'urutan' => 3],
158                 ['jawaban' => 'Keterlibatan pengembangan kurikulum', 'nilai' => 4, 'urutan' => 4],
159                 ['jawaban' => 'Kunjungan industri / company visit', 'nilai' => 5, 'urutan' => 5],
160                 ['jawaban' => 'Penelitian atau proyek bersama', 'nilai' => 6, 'urutan' => 6],
161                 ['jawaban' => 'Kerjasama Corporate Social Responsibility (CSR)', 'nilai' => 7, 'urutan' => 7],
162                 ['jawaban' => 'Riset Pengembangan Produk', 'nilai' => 8, 'urutan' => 8],
163                 ['jawaban' => 'Lainnya', 'nilai' => 9, 'urutan' => 9],
164             ],
165         ],
166     ],
167 ];
168 ];
169
170 DB::beginTransaction();
171
172 try {
173     $now = Carbon::now();
174
175     // Buat record instrumen untuk tahun ini
176     $instrumenId = DB::table('instrumen')->insertGetId([
177         'tahun' => $tahun,
178         'judul' => "Instrumen Evaluasi Pengguna Lulusan {$tahun}",
179         'deskripsi' => 'Draft instrumen evaluasi lulusan oleh pengguna lulusan (perusahaan/instansi).',
180         'is_active' => true,
181         'created_at' => $now,
182         'updated_at' => $now,
183     ]);
184
185     foreach ($instrument as $kategoriNama => $kategoriData) {
186
187         $kategoriId = DB::table('kategoris')->insertGetId([
188             'nama_kategori' => $kategoriNama,
189             'deskripsi' => $kategoriData['deskripsi'],
190             'created_at' => $now,
191             'updated_at' => $now,
192         ]);
193
194         foreach ($kategoriData['soal'] as $kode => $soalData) {
195
196             $soalId = DB::table('soal')->insertGetId([
197                 'instrumen_id' => $instrumenId,
198                 'soal' => $soalData['teks'],
199                 'kode' => $kode,
200                 'kategori_id' => $kategoriId,
201                 'jenis_soal' => $soalData['jenis'],
202                 'peruntukan_fakultas' => 'Umum',

```

```

203         'is_required' => true,
204         'is_active' => true,
205         'created_at' => $now,
206         'updated_at' => $now,
207     ]]);
208
209     if (! empty($soalData['pilihan'])) {
210         $jawabanRows = [];
211         foreach ($soalData['pilihan'] as $p) {
212             $jawabanRows[] = [
213                 'soal_id' => $soalId,
214                 'jawaban' => $p['jawaban'],
215                 'nilai' => $p['nilai'],
216                 'urutan' => $p['urutan'],
217                 'created_at' => $now,
218                 'updated_at' => $now,
219             ];
220         }
221         DB::table('jawaban')->insert($jawabanRows);
222     }
223 }
224 }
225
226 DB::commit();
227 $this->command->info("Instrumen tahun {$tahun} (id={$instrumenId}) berhasil di-seed!");
228
229 } catch (\Exception $e) {
230     DB::rollBack();
231     $this->command->error('Gagal melakukan seeder: '.$e->getMessage());
232     throw $e;
233 }
234 }
235 }
236

```

11. Controller Laporan

File path: app/Http/Controllers/ReportController.php

Controller yang menyiapkan filter laporan, ringkasan survei selesai, daftar arsip, detail arsip, dan proses unduh laporan.

```

1  <?php
2
3  namespace App\Http\Controllers;
4
5  use App\Exports\ReportExport;
6  use App\Models\SurveyArsip;
7  use Illuminate\Http\Request;
8  use Illuminate\Support\Facades\DB;
9
10 class ReportController extends Controller
11 {
12     public function index(Request $request)
13     {
14         $tahunList = DB::table('lulusan')
15             ->whereNotNull('tahun_lulus')
16             ->selectRaw('EXTRACT(YEAR FROM tahun_lulus) as tahun')
17             ->distinct()
18             ->orderByDesc('tahun')
19             ->pluck('tahun');
20
21         $fakultasList = DB::table('lulusan')
22             ->whereNotNull('fakultas')
23             ->distinct()
24             ->orderBy('fakultas')
25             ->pluck('fakultas');
26
27         $prodilist = DB::table('lulusan')
28             ->whereNotNull('program_studi')
29             ->distinct()
30             ->orderBy('program_studi')
31             ->pluck('program_studi');
32
33         $totalSurveySelesai = DB::table('survey')->where('is_completed', true)->count();
34
35         $filters = $request->only(['tahun', 'fakultas', 'program_studi']);
36
37         return view('admin.report.index', compact(
38             'tahunList', 'fakultasList', 'prodilist',
39             'totalSurveySelesai', 'filters'
40         ));
41     }
42
43     public function download(Request $request)
44     {
45         $filters = $request->only(['tahun', 'fakultas', 'program_studi']);
46         return (new ReportExport($filters))->download();

```

```

47     }
48
49     public function arsip(Request $request)
50     {
51         $query = SurveyArsip::query()->orderByDesc('submitted_at');
52
53         if ($request->filled('tahun')) {
54             $query->where('tahun_instrumen', $request->tahun);
55         }
56         if ($request->filled('fakultas')) {
57             $query->where('lulusan_fakultas', $request->fakultas);
58         }
59         if ($request->filled('program_studi')) {
60             $query->where('lulusan_program_studi', $request->program_studi);
61         }
62         if ($request->filled('cari')) {
63             $cari = $request->cari;
64             $query->where(function ($q) use ($cari) {
65                 $q->where('lulusan_nama', 'like', "%{$cari}%")
66                     ->orWhere('lulusan_nim', 'like', "%{$cari}%")
67                     ->orWhere('perusahaan_nama', 'like', "%{$cari}%")
68                     ->orWhere('penyelia_nama', 'like', "%{$cari}%");
69             });
70         }
71
72         $arsip = $query->paginate(20)->withQueryString();
73
74         $tahunList = SurveyArsip::whereNotNull('tahun_instrumen')->distinct()->orderByDesc('tahun_instrumen')->pluck('tahun_instrumen');
75         $fakultasList = SurveyArsip::whereNotNull('lulusan_fakultas')->distinct()->orderBy('lulusan_fakultas')->pluck('lulusan_fakultas');
76         $prodilist = SurveyArsip::whereNotNull('lulusan_program_studi')->distinct()->orderBy('lulusan_program_studi')->pluck('lulusan_program_studi');
77
78         return view('admin.report.arsip', compact('arsip', 'tahunList', 'fakultasList', 'prodilist'));
79     }
80
81     public function arsipDetail($id)
82     {
83         $arsip = SurveyArsip::findOrFail($id);
84         return view('admin.report.arsip-detail', compact('arsip'));
85     }
86 }
87

```

12. Ekspor dan Perhitungan Laporan

File path: app/Exports/ReportExport.php

Generator laporan spreadsheet yang menghitung distribusi persentase jawaban, subtotal kategori, total, dan rata-rata per program studi.

```

1  <?php
2
3  namespace App\Exports;
4
5  use Illuminate\Support\Facades\DB;
6  use PhpOffice\PhpSpreadsheet\Spreadsheet;
7  use PhpOffice\PhpSpreadsheet\Style\Alignment;
8  use PhpOffice\PhpSpreadsheet\Style\Border;
9  use PhpOffice\PhpSpreadsheet\Style\Fill;
10 use PhpOffice\PhpSpreadsheet\Writer\Xlsx;
11
12 class ReportExport
13 {
14     private array $filters;
15
16     // — Palet Warna Global —————
17     private const TITLE_BG    = 'FF1E3A5F'; // Navy gelap - judul utama
18     private const SECTION_BG  = 'FF2D3748'; // Biru-abu gelap - divider bagian
19     private const ROW_ALT     = 'FFF7F9FC'; // Abu sangat muda - baris genap
20     private const BORDER      = 'FFD0D7DE'; // Abu muda - border
21     private const INFO_BG     = 'FFECF1F7'; // Abu-biru - blok info prodi
22     private const INFO_LBL    = 'FF4A5568'; // Abu-gelap - label info
23     private const SUBTOTAL_BG = 'FFFFD966'; // Amber keemasan - baris subtotal kategori
24     private const TOTAL_BG    = 'FFE2E8F0'; // Abu-biru muda - baris total
25     private const RATA_BG     = 'FFCBD5E1'; // Abu-biru - baris rata-rata
26
27     // Warna header kolom rating (semantik)
28     private const RATING_HDR = [
29         4 => 'FF1E7E34', // Hijau tua - Sangat Baik
30         3 => 'FF0B5ED7', // Biru tua - Baik
31         2 => 'FFD4690C', // Oranye tua - Kurang
32         1 => 'FFBB2D3B', // Merah tua - Sangat Kurang
33     ];
34
35     // Warna aksen per fakultas
36     private array $fakultasColors = [
37         'FTI' => 'FF1B4F8A', // Biru navy
38         'FDIK' => 'FF7D3C00', // Coklat tua
39         'FEB' => 'FF1E5E2E', // Hijau tua
40     ];
41
42     // Warna header data-table (lebih terang dari aksen)
43     private array $fakultasHeaderLight = [
44         'FTI' => 'FF2E75B6',
45         'FDIK' => 'FFA05000',
46         'FEB' => 'FF2D7A43',

```

```

47 ];
48
49 private array $nilaiLabel = [
50     4 => 'Sangat Baik (4)',
51     3 => 'Baik (3)',
52     2 => 'Kurang (2)',
53     1 => 'Sangat Kurang (1)',
54 ];
55
56 public function __construct(array $filters)
57 {
58     $this->filters = $filters;
59 }
60
61 public function build(): Spreadsheet
62 {
63     $spreadsheet = new Spreadsheet();
64     $spreadsheet->removeSheetByIndex(0);
65
66     $tahunList = $this->getTahunList();
67
68     if (empty($tahunList)) {
69         $sheet = $spreadsheet->createSheet();
70         $sheet->setTitle('Tidak Ada Data');
71         $sheet->setCellValue('A1', 'Tidak ada data survey yang selesai sesuai filter yang dipilih.');
```

```

72         return $spreadsheet;
73     }
74
75     foreach ($tahunList as $tahun) {
76         $sheet = $spreadsheet->createSheet();
77         $sheet->setTitle((string) $tahun);
78         $this->buildSheet($sheet, $tahun);
79     }
80
81     return $spreadsheet;
82 }
83
84 private function buildSheet(\PhpOffice\PhpSpreadsheet\Worksheet\Worksheet $sheet, int $tahun): void
85 {
86     $row = 1;
87
88     // — Judul Utama —————
89     $sheet->mergeCells("A{$row}:R{$row}");
90     $sheet->setCellValue("A{$row}", 'LAPORAN TRACER STUDY – EVALUASI PENGGUNA LULUSAN');
```

```

91     $sheet->getStyle("A{$row}")->applyFromArray([
92         'font'      => ['bold' => true, 'size' => 14, 'color' => ['argb' => 'FFFFFF']],
93         'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => self::TITLE_BG]],
94         'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER, 'vertical' => Alignment::VERTICAL_CENTER],
95     ]);
96     $sheet->getRowDimension($row)->setRowHeight(28);
97     $row++;
98
99     $sheet->mergeCells("A{$row}:R{$row}");
```

```

100 $sheet->setCellValue("A{$row}", 'Universitas Dinamika');
101 $sheet->getStyle("A{$row}")->applyFromArray([
102     'font'      => ['bold' => true, 'size' => 11, 'color' => ['argb' => 'FFFFFFF']],
103     'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => self::TITLE_BG]],
104     'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER],
105 ]);
106 $row++;
107
108 $filterText = "Tahun Lulus: {$tahun}"
109     . ($this->filters['fakultas'] ? ' | Fakultas: ' . $this->filters['fakultas'] : '')
110     . ($this->filters['program_studi'] ? ' | Prodi: ' . $this->filters['program_studi'] : '');
111 $sheet->mergeCells("A{$row}:R{$row}");
112 $sheet->setCellValue("A{$row}", $filterText);
113 $sheet->getStyle("A{$row}")->applyFromArray([
114     'font'      => ['italic' => true, 'size' => 9, 'color' => ['argb' => 'FFAABCCE']],
115     'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => self::TITLE_BG]],
116     'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER],
117 ]);
118 $row++;
119 $row++; // baris kosong
120
121 $fakultasList = $this->filters['fakultas']
122     ? [$this->filters['fakultas']]
123     : ['FTI', 'FDIK', 'FEB'];
124
125 $summaryDataPerFakultas = [];
126
127 foreach ($fakultasList as $fakultas) {
128     $soalList = $this->getSoalForFakultas($fakultas);
129     $dataRows = $this->getDataRows($tahun, $fakultas);
130
131     if ($dataRows->isEmpty()) {
132         continue;
133     }
134
135     $color      = $this->fakultasColors[$fakultas] ?? 'FF1B4F8A';
136     $colorLight = $this->fakultasHeaderLight[$fakultas] ?? 'FF2E75B6';
137     $colEnd     = $this->colLetter(9 + count($soalList) - 1);
138
139     // — Header Fakultas —————
140     $sheet->mergeCells("A{$row}:{$colEnd}{$row}");
141     $sheet->setCellValue("A{$row}", "FAKULTAS: {$fakultas}");
142     $sheet->getStyle("A{$row}:{$colEnd}{$row}")->applyFromArray([
143         'font'      => ['bold' => true, 'size' => 11, 'color' => ['argb' => 'FFFFFFF']],
144         'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => $color]],
145         'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER, 'vertical' => Alignment::VERTICAL_CENTER],
146     ]);
147     $sheet->getRowDimension($row)->setRowHeight(22);
148     $row++;
149
150     // — Header Kolom Data —————
151     $headers = ['No', 'Prodi', 'NIM', 'Nama Alumni', 'Nama Responden', 'Nama Perusahaan', 'Jenis Perusahaan', 'Cab. Kota', 'Cab. Negara'];
152     foreach ($soalList as $soal) {

```

```

153     $headers[] = $soal->kode;
154 }
155
156 $col = 'A';
157 foreach ($headers as $header) {
158     $sheet->setCellValue("{ $col}{$row}", $header);
159     $col++;
160 }
161 $sheet->getStyle("A{$row}:{ $colEnd}{$row}")->applyFromArray([
162     'font'      => ['bold' => true, 'color' => ['argb' => 'FFFFFFF']],
163     'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => $colorLight]],
164     'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER, 'vertical' => Alignment::VERTICAL_CENTER, 'wrapText' => true],
165     'borders'   => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
166 ]);
167 $sheet->getRowDimension($row)->setRowHeight(25);
168 $row++;
169
170 // — Baris Data —————
171 $no      = 1;
172 $soalIdList = $soalList->pluck('id')->toArray();
173 $peProdi  = [];
174
175 foreach ($dataRows as $lulusan) {
176     $col = 'A';
177     $sheet->setCellValue("{ $col}{$row}", $no++); $col++;
178     $sheet->setCellValue("{ $col}{$row}", $lulusan->program_studi); $col++;
179     $sheet->setCellValue("{ $col}{$row}", $lulusan->nim); $col++;
180     $sheet->setCellValue("{ $col}{$row}", $lulusan->nama); $col++;
181     $sheet->setCellValue("{ $col}{$row}", $lulusan->nama_penyelida); $col++;
182     $sheet->setCellValue("{ $col}{$row}", $lulusan->nama_perusahaan); $col++;
183     $sheet->setCellValue("{ $col}{$row}", ucfirst($lulusan->jenis_perusahaan ?? '-')); $col++;
184     $sheet->setCellValue("{ $col}{$row}", $lulusan->cabang_kota ? 'Ya' : 'Tidak'); $col++;
185     $sheet->setCellValue("{ $col}{$row}", $lulusan->cabang_negara ? 'Ya' : 'Tidak'); $col++;
186
187     $jawabanMap = $this->getJawabanForSurvey($lulusan->survey_id, $soalIdList);
188
189     foreach ($soalList as $soal) {
190         $nilai = $jawabanMap[$soal->id] ?? null;
191         $label = $nilai ? ($this->nilaiLabel[$nilai] ?? $nilai) : '-';
192         $sheet->setCellValue("{ $col}{$row}", $label);
193         $col++;
194     }
195
196     $bgColor = ($no % 2 === 0) ? self::ROW_ALT : 'FFFFFFF';
197     $sheet->getStyle("A{$row}:{ $colEnd}{$row}")->applyFromArray([
198         'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => $bgColor]],
199         'borders'   => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
200         'alignment' => ['vertical' => Alignment::VERTICAL_CENTER],
201     ]);
202
203     $peProdi[$lulusan->program_studi][] = $jawabanMap;
204     $row++;
205 }

```



```

206
207     $summaryDataPerFakultas[$fakultas] = [
208         'soallist' => $soallist,
209         'peProdi'  => $peProdi,
210         'color'    => $color,
211         'colorLight' => $colorLight,
212     ];
213
214     $row += 2;
215 }
216
217 // — Tabel Ringkasan —————
218 if (empty($summaryDataPerFakultas)) {
219     return;
220 }
221
222 $sheet->mergeCells("A{$row}:E{$row}");
223 $sheet->setCellValue("A{$row}", 'RINGKASAN DISTRIBUSI PENILAIAN PER PROGRAM STUDI');
224 $sheet->getStyle("A{$row}:E{$row}")->applyFromArray([
225     'font'      => ['bold' => true, 'size' => 12, 'color' => ['argb' => 'FFFFFF']],
226     'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => self::SECTION_BG]],
227     'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER, 'vertical' => Alignment::VERTICAL_CENTER],
228 ]);
229 $sheet->getRowDimension($row)->setRowHeight(22);
230 $row++;
231 $row++;
232
233 foreach ($summaryDataPerFakultas as $fakultas => $sData) {
234     $soallist = $sData['soallist'];
235     $peProdi  = $sData['peProdi'];
236     $color    = $sData['color'];
237     $colorLight = $sData['colorLight'];
238
239     foreach ($peProdi as $prodi => $jawabanRows) {
240         $jumlahResp = count($jawabanRows);
241
242         // — Blok Info Prodi —————
243         $infoItems = [
244             'Prodi'      => $prodi,
245             'Fakultas'   => $fakultas,
246             'Tahun Lulus' => $tahun,
247             'Jumlah Resp' => $jumlahResp,
248         ];
249         foreach ($infoItems as $label => $value) {
250             $sheet->setCellValue("A{$row}", $label);
251             $sheet->setCellValue("B{$row}", $value);
252             $sheet->getStyle("A{$row}")->applyFromArray([
253                 'font' => ['bold' => true, 'color' => ['argb' => self::INFO_LBL]],
254                 'fill' => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => self::INFO_BG]],
255                 'borders' => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
256             ]);
257             $sheet->getStyle("B{$row}")->applyFromArray([
258                 'font' => ['bold' => ($label === 'Prodi')],

```

```

259         'borders' => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
260     ]);
261     $row++;
262 }
263 $row++; // gap
264
265 // — Header Distribusi per Soal —————
266 // Kolom A: nama soal (warna aksen fakultas)
267 $sheet->setCellValue("A{$row}", 'Kode Soal');
268 $sheet->getStyle("A{$row}")->applyFromArray([
269     'font'      => ['bold' => true, 'color' => ['argb' => 'FFFFFF']],
270     'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => $colorLight]],
271     'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER],
272     'borders'   => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
273 ]);
274
275 // Kolom B-E: warna semantik rating
276 $ratingLabels = [4 => 'Sangat Baik (4)', 3 => 'Baik (3)', 2 => 'Kurang (2)', 1 => 'Sangat Kurang (1)'];
277 $ratingCols   = [4 => 'B', 3 => 'C', 2 => 'D', 1 => 'E'];
278 foreach ($ratingLabels as $rating => $label) {
279     $rCol = $ratingCols[$rating];
280     $sheet->setCellValue("{$rCol}{$row}", $label);
281     $sheet->getStyle("{$rCol}{$row}")->applyFromArray([
282         'font'      => ['bold' => true, 'color' => ['argb' => 'FFFFFF']],
283         'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => self::RATING_HDR[$rating]]],
284         'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER],
285         'borders'   => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
286     ]);
287 }
288 $row++;
289
290 // — Soal per Kategori + Subtotal Kategori —————
291 $soalPerKategori = [];
292 foreach ($soalList as $soal) {
293     $key = $soal->nama_kategori ?? 'Tidak Berkategori';
294     $soalPerKategori[$key][] = $soal;
295 }
296
297 $kategoriDistribusi = [];
298 $soalNo = 0;
299
300 foreach ($soalPerKategori as $namaKategori => $soalDalamKategori) {
301     $kategoriAccum = [4 => [], 3 => [], 2 => [], 1 => []];
302
303     foreach ($soalDalamKategori as $soal) {
304         $counts = [4 => 0, 3 => 0, 2 => 0, 1 => 0];
305         foreach ($jawabanRows as $jawabanMap) {
306             $nilai = $jawabanMap[$soal->id] ?? null;
307             if ($nilai && isset($counts[$nilai])) {
308                 $counts[$nilai]++;
309             }
310         }
311         $total = array_sum($counts);

```

```

312
313     $pctSB = $total > 0 ? round($counts[4] / $total * 100, 1) : 0;
314     $pctB  = $total > 0 ? round($counts[3] / $total * 100, 1) : 0;
315     $pctK  = $total > 0 ? round($counts[2] / $total * 100, 1) : 0;
316     $pctSK = $total > 0 ? round($counts[1] / $total * 100, 1) : 0;
317
318     $soalNo++;
319     $rowBg = ($soalNo % 2 === 0) ? self::ROW_ALT : 'FFFFFFF';
320
321     $sheet->setCellValue("A{$row}", $soal->kode);
322     $sheet->setCellValue("B{$row}", $pctSB . '%');
323     $sheet->setCellValue("C{$row}", $pctB . '%');
324     $sheet->setCellValue("D{$row}", $pctK . '%');
325     $sheet->setCellValue("E{$row}", $pctSK . '%');
326     $sheet->getStyle("A{$row}:E{$row}")->applyFromArray([
327         'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => $rowBg]],
328         'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER],
329         'borders'   => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
330     ]);
331     $sheet->getStyle("A{$row}")->getAlignment()->setHorizontal(Alignment::HORIZONTAL_LEFT);
332
333     $kategoriAccum[4][] = $pctSB;
334     $kategoriAccum[3][] = $pctB;
335     $kategoriAccum[2][] = $pctK;
336     $kategoriAccum[1][] = $pctSK;
337     $row++;
338 }
339
340 // Baris subtotal kategori (amber)
341 $avgSB = count($kategoriAccum[4]) > 0 ? round(array_sum($kategoriAccum[4]) / count($kategoriAccum[4]), 1) : 0;
342 $avgB  = count($kategoriAccum[3]) > 0 ? round(array_sum($kategoriAccum[3]) / count($kategoriAccum[3]), 1) : 0;
343 $avgK  = count($kategoriAccum[2]) > 0 ? round(array_sum($kategoriAccum[2]) / count($kategoriAccum[2]), 1) : 0;
344 $avgSK = count($kategoriAccum[1]) > 0 ? round(array_sum($kategoriAccum[1]) / count($kategoriAccum[1]), 1) : 0;
345
346 $sheet->setCellValue("A{$row}", $namaKategori);
347 $sheet->setCellValue("B{$row}", $avgSB . '%');
348 $sheet->setCellValue("C{$row}", $avgB . '%');
349 $sheet->setCellValue("D{$row}", $avgK . '%');
350 $sheet->setCellValue("E{$row}", $avgSK . '%');
351 $sheet->getStyle("A{$row}:E{$row}")->applyFromArray([
352     'font'      => ['bold' => true, 'color' => ['argb' => 'FF1A2637']],
353     'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => self::SUBTOTAL_BG]],
354     'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER],
355     'borders'   => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => 'FFCCAA00']]],
356 ]);
357 $sheet->getStyle("A{$row}")->getAlignment()->setHorizontal(Alignment::HORIZONTAL_LEFT);
358
359 $kategoriDistribusi[$namaKategori] = [4 => $avgSB, 3 => $avgB, 2 => $avgK, 1 => $avgSK];
360 $row++;
361 }
362
363 // — Tabel Ringkasan Tingkat Kepuasan per Kategori —————
364 $row++;

```

```

365 $sheet->mergeCells("A{$row}:E{$row}");
366 $sheet->setCellValue("A{$row}", 'TINGKAT KEPUASAN PENGGUNA (%)');
367 $sheet->getStyle("A{$row}:E{$row}")->applyFromArray([
368     'font'      => ['bold' => true, 'size' => 11, 'color' => ['argb' => 'FFFFFFF']],
369     'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => $color]],
370     'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER, 'vertical' => Alignment::VERTICAL_CENTER],
371 ]);
372 $sheet->getRowDimension($row->setRowHeight(20));
373 $row++;
374
375 // Header kolom summary – tetap warna semantik per rating
376 $sheet->setCellValue("A{$row}", 'Jenis Kemampuan');
377 $sheet->getStyle("A{$row}")->applyFromArray([
378     'font'      => ['bold' => true, 'color' => ['argb' => 'FFFFFFF']],
379     'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => $colorLight]],
380     'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER],
381     'borders'   => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
382 ]);
383 foreach ($ratingLabels as $rating => $label) {
384     $rCol = $ratingCols[$rating];
385     $sheet->setCellValue("${$rCol}{$row}", $label);
386     $sheet->getStyle("${$rCol}{$row}")->applyFromArray([
387         'font'      => ['bold' => true, 'color' => ['argb' => 'FFFFFFF']],
388         'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => self::RATING_HDR[$rating]]],
389         'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER],
390         'borders'   => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
391     ]);
392 }
393 $row++;
394
395 $totalSB = 0; $totalB = 0; $totalK = 0; $totalSK = 0;
396 $katNo = 0;
397 foreach ($kategoriDistribusi as $namaKategori => $pcts) {
398     $katNo++;
399     $rowBg = ($katNo % 2 === 0) ? self::ROW_ALT : 'FFFFFFF';
400     $sheet->setCellValue("A{$row}", $namaKategori);
401     $sheet->setCellValue("B{$row}", $pcts[4] . '%');
402     $sheet->setCellValue("C{$row}", $pcts[3] . '%');
403     $sheet->setCellValue("D{$row}", $pcts[2] . '%');
404     $sheet->setCellValue("E{$row}", $pcts[1] . '%');
405     $sheet->getStyle("A{$row}:E{$row}")->applyFromArray([
406         'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => $rowBg]],
407         'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER],
408         'borders'   => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
409     ]);
410     $sheet->getStyle("A{$row}")->getAlignment()->setHorizontal(Alignment::HORIZONTAL_LEFT);
411     $totalSB += $pcts[4];
412     $totalB += $pcts[3];
413     $totalK += $pcts[2];
414     $totalSK += $pcts[1];
415     $row++;
416 }
417

```

```

418 // Baris Total
419 $sheet->setCellValue("A{$row}", 'Total');
420 $sheet->setCellValue("B{$row}", round($totalSB, 1) . '%');
421 $sheet->setCellValue("C{$row}", round($totalB, 1) . '%');
422 $sheet->setCellValue("D{$row}", round($totalK, 1) . '%');
423 $sheet->setCellValue("E{$row}", round($totalSK, 1) . '%');
424 $sheet->getStyle("A{$row}:E{$row}")->applyFromArray([
425     'font'      => ['bold' => true, 'color' => ['argb' => 'FF1A2637']],
426     'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => self::TOTAL_BG]],
427     'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER],
428     'borders'   => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
429 ]);
430 $row++;
431
432 // Baris Rata-Rata
433 $countKat = count($kategoriDistribusi);
434 $sheet->setCellValue("A{$row}", 'Rata - Rata');
435 $sheet->setCellValue("B{$row}", ($countKat > 0 ? round($totalSB / $countKat, 1) : 0) . '%');
436 $sheet->setCellValue("C{$row}", ($countKat > 0 ? round($totalB / $countKat, 1) : 0) . '%');
437 $sheet->setCellValue("D{$row}", ($countKat > 0 ? round($totalK / $countKat, 1) : 0) . '%');
438 $sheet->setCellValue("E{$row}", ($countKat > 0 ? round($totalSK / $countKat, 1) : 0) . '%');
439 $sheet->getStyle("A{$row}:E{$row}")->applyFromArray([
440     'font'      => ['bold' => true, 'color' => ['argb' => 'FF1A2637']],
441     'fill'      => ['fillType' => Fill::FILL_SOLID, 'startColor' => ['argb' => self::RATA_BG]],
442     'alignment' => ['horizontal' => Alignment::HORIZONTAL_CENTER],
443     'borders'   => ['allBorders' => ['borderStyle' => Border::BORDER_THIN, 'color' => ['argb' => self::BORDER]]],
444 ]);
445 $row++;
446
447 $row += 2;
448 }
449 }
450
451 foreach (range('A', 'I') as $c) {
452     $sheet->getColumnDimension($c)->setAutoSize(true);
453 }
454 $sheet->getColumnDimension('E')->setAutoSize(true);
455 }
456
457 private function getTahunList(): array
458 {
459     $q = DB::table('lulusan')
460         ->join('survey', 'lulusan.id', '=', 'survey.lulusan_id')
461         ->where('survey.is_completed', true)
462         ->selectRaw('EXTRACT(YEAR FROM lulusan.tahun_lulus) as tahun')
463         ->distinct();
464
465     if ($this->filters['tahun']) {
466         $q->whereRaw('EXTRACT(YEAR FROM lulusan.tahun_lulus) = ?', [$this->filters['tahun']]);
467     }
468     if ($this->filters['fakultas']) {
469         $q->where('lulusan.fakultas', $this->filters['fakultas']);
470     }

```

```

471     if ($this->filters['program_studi']) {
472         $q->where('lulusan.program_studi', $this->filters['program_studi']);
473     }
474
475     return $q->orderBy('tahun')->pluck('tahun')->map(fn($v) => (int) $v)->toArray();
476 }
477
478 private function getSoalForFakultas(string $fakultas)
479 {
480     return DB::table('soal')
481         ->leftJoin('kategoris', 'soal.kategori_id', '=', 'kategoris.id')
482         ->where('soal.is_active', true)
483         ->where('soal.jenis_soal', 'rating')
484         ->where(function ($q) use ($fakultas) {
485             $q->where('soal.peruntukan_fakultas', 'Umum')
486                 ->orWhere('soal.peruntukan_fakultas', $fakultas);
487         })
488         ->orderBy('soal.kode')
489         ->get(['soal.id', 'soal.kode', 'soal.soal', 'soal.kategori_id', 'kategoris.nama_kategori']);
490 }
491
492 private function getDataRows(int $tahun, string $fakultas)
493 {
494     $q = DB::table('survey')
495         ->join('lulusan', 'survey.lulusan_id', '=', 'lulusan.id')
496         ->join('pengguna_lulusan', 'survey.pengguna_lulusan_id', '=', 'pengguna_lulusan.id')
497         ->where('survey.is_completed', true)
498         ->where('lulusan.fakultas', $fakultas)
499         ->whereRaw('EXTRACT(YEAR FROM lulusan.tahun_lulus) = ?', [$tahun])
500         ->select(
501             'survey.id as survey_id',
502             'lulusan.nama',
503             'lulusan.nim',
504             'lulusan.program_studi',
505             'pengguna_lulusan.nama_penyelid',
506             'pengguna_lulusan.nama_perusahaan',
507             'pengguna_lulusan.jenis_perusahaan',
508             'pengguna_lulusan.cabang_kota',
509             'pengguna_lulusan.cabang_negara'
510         )
511         ->orderBy('lulusan.program_studi')
512         ->orderBy('lulusan.nama');
513
514     if ($this->filters['program_studi']) {
515         $q->where('lulusan.program_studi', $this->filters['program_studi']);
516     }
517
518     return $q->get();
519 }
520
521 private function getJawabanForSurvey(int $surveyId, array $soalIds): array
522 {
523     $rows = DB::table('respon_jawaban')

```

```

524     ->join('jawaban', 'respon_jawaban.jawaban_id', '=', 'jawaban.id')
525     ->where('respon_jawaban.survey_id', $surveyId)
526     ->whereIn('respon_jawaban.soal_id', $soalIds)
527     ->select('respon_jawaban.soal_id', 'jawaban.nilai')
528     ->get();
529
530     $map = [];
531     foreach ($rows as $r) {
532         $map[$r->soal_id] = (int) $r->nilai;
533     }
534     return $map;
535 }
536
537 private function colLetter(int $n): string
538 {
539     return \PhpOffice\PhpSpreadsheet\Cell\Coordinate::stringFromColumnIndex($n);
540 }
541
542 public function download(): \Symfony\Component\HttpFoundation\StreamedResponse
543 {
544     $spreadsheet = $this->build();
545
546     $tahun    = $this->filters['tahun']      ?? 'Semua';
547     $fak      = $this->filters['fakultas']    ?? 'Semua';
548     $prodi    = $this->filters['program_studi'] ?? '';
549     $parts    = array_filter(["Tahun{$tahun}", "Fak{$fak}", $prodi ? 'Prodi_' . str_replace(' ', '_', $prodi) : '']);
550     $filename = 'Laporan_Tracer_Study_' . implode('_', $parts) . '.xlsx';
551
552     $writer = new Xlsx($spreadsheet);
553
554     return response()->streamDownload(function () use ($writer) {
555         $writer->save('php://output');
556     }, $filename, [
557         'Content-Type'      => 'application/vnd.openxmlformats-officedocument.spreadsheetml.sheet',
558         'Content-Disposition' => 'attachment; filename="' . $filename . '"',
559         'Cache-Control'     => 'max-age=0',
560     ]);
561 }
562 }
563

```

13. Model Survei

File path: app/Models/Survey.php

Model utama survei beserta relasi lulusan, pengguna lulusan, pertanyaan, instrumen, respon jawaban, dan arsip.

```
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class Survey extends Model
8 {
9     protected $table = 'survey';
10
11     protected $fillable = [
12         'access_code',
13         'lulusan_id',
14         'pengguna_lulusan_id',
15         'judul',
16         'tahun',
17         'deskripsi',
18         'is_completed',
19         'is_active',
20     ];
21
22     /**
23      * Relasi ke tabel lulusan
24      */
25     public function soals()
26     {
27         return $this->belongsToMany(Soal::class, 'survey_soal', 'survey_id', 'soal_id');
28     }
29     public function lulusan()
30     {
31         return $this->belongsTo(Lulusan::class);
32     }
33
34     /**
35      * Relasi ke pengguna lulusan (perusahaan)
36      */
37     public function penggunaLulusan()
38     {
39         return $this->belongsTo(PenggunaLulusan::class);
40     }
41
42     /**
43      * Relasi ke jawaban (jika masih dipakai)
44      */
45     public function jawaban()
46     {
```



```
47         return $this->hasMany(Jawaban::class);  
48     }  
49 }  
50
```

14. Model Arsip Survei

File path: app/Models/SurveyArsip.php

Model snapshot permanen hasil survei yang menjadi sumber data stabil untuk dashboard dan pelaporan.

```
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class SurveyArsip extends Model
8 {
9     protected $table = 'survey_arsip';
10
11     protected $fillable = [
12         'survey_id',
13         'access_code',
14         'judul',
15         'submitted_at',
16         'tahun_instrumen',
17
18         'lulusan_nama',
19         'lulusan_nim',
20         'lulusan_program_studi',
21         'lulusan_fakultas',
22         'lulusan_tahun_lulus',
23
24         'perusahaan_nama',
25         'perusahaan_jenis',
26         'perusahaan_alamat',
27         'perusahaan_kontak',
28         'perusahaan_nomor_badan_hukum',
29         'perusahaan_cabang_kota',
30         'perusahaan_cabang_negara',
31
32         'penyelia_nama',
33         'penyelia_jabatan',
34         'penyelia_email',
35         'penyelia_kontak',
36         'jumlah_lulusan_bekerja',
37
38         'jawaban_json',
39     ];
40
41     protected $casts = [
42         'submitted_at' => 'datetime',
43         'jawaban_json' => 'array',
44     ];
45 }
46
```


15. Model Respon Jawaban

File path: app/Models/ResponJawaban.php

Model detail jawaban responden yang menyimpan relasi soal, pilihan jawaban, nilai, dan teks snapshot.

```
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class ResponJawaban extends Model
8 {
9     protected $table = 'respon_jawaban';
10
11     protected $fillable = [
12         'survey_id',
13         'soal_id',
14         'soal_text_snapshot',
15         'jawaban_id',
16         'jawaban_text_snapshot',
17         'jawaban_text',
18         'responden',
19         'jumlah_lulus_bekerja',
20     ];
21
22     // Relasi ke Survey (sesi pengisian)
23     public function survey()
24     {
25         return $this->belongsTo(Survey::class);
26     }
27
28     // Relasi ke Soal (pertanyaan)
29     public function soal()
30     {
31         return $this->belongsTo(Soal::class);
32     }
33
34     // Relasi ke Jawaban (opsi pilihan)
35     public function jawaban()
36     {
37         return $this->belongsTo(Jawaban::class);
38     }
39 }
40
```